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Fundamentals of Upper Urinary Tract Drainage

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Keywords

nephrostomy tube; upper urinary tract drainage; ureteral stent

Questions

1. Percutaneous nephrostomy is not indicated for:
    a. instillation of intracavitary topical therapy for urothelial carcinoma.
    b. Whitaker test.
    c. management of fungal bezoars.
    d. urinary retention.
    e. ureteral injury.
2. Relative to retrograde ureteral stent placement, percutaneous nephrostomy:
    a. has a lower success rate.
    b. requires less anesthesia.
    c. is preferred in cases of ureteral obstruction owing to malignancy.
    d. is less commonly complicated by bacteriuria after indwelling for 1 week.
    e. is associated with worse health-related quality-of-life scores.

3. Which of the following is correct regarding the orientation of the kidney?
    a. The right kidney is slightly cephalad to the left kidney.
    b. The longitudinal axis is 45 degrees from vertical, with the lower pole lateral to the upper pole.
    c. The longitudinal axis is 45 degrees from vertical, with the lower pole anterior to the upper pole.
    d. The apposition of the colon to the kidney is greatest on the left side at the upper pole.
    e. Immediately posterior to the kidneys are the quadratus lumborum muscle, the psoas muscle, and the diaphragm.
4. Which of the following is correct regarding the intrarenal collecting system?
    a. Paired anterior and posterior calyces enter the infundibula approximately 90 degrees from each other.
    b. Compound calyces are most common in the lower pole
    c. Most kidneys have three distinct infundibula: the upper, middle, and lower.
    d. There are 8 to 16 minor calyces.
    e. There is a consistent relationship between anterior and posterior calyces and their medial-lateral position on anterior-posterior radiography.
5. The correct order of the division of the intrarenal branches of the renal artery is:
    a. segmental, arcuate, interlobar (infundibular), interlobular.
    b. segmental, arcuate, interlobular, interlobar (infundibular).
    c. segmental, interlobar (infundibular), arcuate, interlobular.
    d. interlobular, segmental, interlobar (infundibular), arcuate.
    e. segmental, interlobular, interlobar (infundibular), arcuate.
6. To reduce the risk of infectious complications from percutaneous renal surgery:
    a. all patients should receive prophylactic antimicrobials.
    b. urine cultures should be obtained on all patients.
    c. urine must be sterile before the procedure.
    d. gentamicin is an acceptable single agent for antimicrobial prophylaxis.

e. ampicillin/sulbactam is not an acceptable single agent for antimicrobial prophylaxis.

7. To reduce the risk of hemorrhagic complications associated with percutaneous renal access, the minimum recommended preoperative cessation period for:
   a. herbal medications is 2 weeks.
   b. clopidogrel is 10 days.
   c. aspirin is 5 days.
   d. warfarin is 5 days.
   e. nonsteroidal inflammatory agents is 1 day.
8. Which of the following have NOT been demonstrated in randomized controlled clinical trials to reduce pain associated with percutaneous renal access?
   a. Tract infiltration with local anesthetic
   b. Intercostal nerve block
   c. Thoracic paravertebral block
   d. Balloon dilation compared with semirigid plastic dilation of the access tract
   e. Smaller, compared with larger, caliber postprocedure nephrostomy tubes
9. An advantage of the supine versus prone position for percutaneous renal surgery is:
   a. improved pulmonary mechanics.
   b. a large horizontal working surface.
   c. easier entry into upper pole calyces.
   d. easier entry into posterior calyces.
   e. reduced pressure in the collecting system.
10. Access into which site provides the optimal versatility and safety for percutaneous renal surgery in the prone position?
    a. Upper pole posterior calyx
    b. Upper pole infundibulum
    c. Renal pelvis
    d. Middle calyx
    e. Lower pole anterior calyx
11. Techniques for retrograde assistance for percutaneous renal access include all but which of the following?

a. Straight ureteral catheter to inject air
b. Ureteral access sheath to facilitate drainage
c. Ureteroscopy to retrieve guidewire
d. Retrograde approach to percutaneous access
e. Retrograde placement of externalized (single pigtail) ureteral stent for drainage

12. Compared with an 18-gauge needle, the 21-gauge needle for percutaneous renal access:
    a. should not be used by inexperienced operators.
    b. requires a 0.025-inch guidewire.
    c. cannot be directed as easily.
    d. entails less risk of loss of access.
    e. is more traumatic.
13. Compared with ultrasonography, fluoroscopy for percutaneous renal access:
    a. is less suited in the morbidly obese.
    b. provides more rapid evaluation of the entire kidney.
    c. cannot be used to monitor tract dilation.
    d. visualizes the access needle better.
    e. is preferred in transplant kidneys.
14. The "triangulation" technique for fluoroscopic percutaneous renal access:
    a. increases radiation exposure to the operator's hands compared with the "eye-of-the-needle" technique.
    b. cannot be performed in malrotated kidneys.
    c. is not as dependent on retrograde assistance as the "eye-of-the-needle" technique.
    d. is less suitable than the "eye-of-the-needle" technique in morbidly obese patients.
    e. continuously monitors depth of needle penetration.
15. Dilation of the tract for percutaneous renal surgery is:
    a. not effective with a balloon dilator in hypermobile kidneys.
    b. most effective with semirigid dilator.
    c. least expensive with metal dilators.
    d. most rapid with metal dilators.
    e. easiest with a one-shot semirigid dilator.

16. When considering percutaneous renal surgery in horseshoe kidneys:
    a. upper pole access is dangerous.
    b. lower pole access is preferred in most cases.
    c. computed tomography can be misleading.
    d. the puncture site is more lateral than in normal kidneys.
    e. lower hemorrhage rates than in normal kidneys can be expected.
17. When considering percutaneous renal surgery in transplant kidneys:
    a. retrograde assistance is difficult.
    b. fluoroscopy is more useful than ultrasonography for initial access.
    c. the typical hypermobility renders tract dilation difficult.
    d. semirigid plastic dilators should not be used.
    e. secondary procedures are usually required.
18. Foley catheters for postprocedure nephrostomy drainage:
    a. do not need to be secured at the skin.
    b. can have a ureteral catheter passed through the end.
    c. should have the balloon filled with dilute contrast material.
    d. stay more securely in the kidney than Malecot catheters.
    e. are less likely to become infected than Malecot catheters.
19. The Cope retention mechanism:
    a. is used in nephroureteral stents.
    b. is used in internal ureteral stents.
    c. is more secure than a balloon catheter.
    d. requires cutting the tube to disengage.
    e. should not be used in more than one access site.
20. Alternatives to a nephrostomy tube after percutaneous renal surgery include all EXCEPT:
    a. maintenance of the working sheath.
    b. an internal ureteral stent that is removed cystoscopically.
    c. an internal ureteral stent with an attached string that exits out the flank.
    d. a ureteral stent externalized out the urethra.
    e. no drainage tube at all.
21. A postoperative nephrostomy tube:

a. offers greater assurance of upper urinary tract drainage than an internal ureteral stent.
b. should be placed in the dilated access site.
c. does not maintain the percutaneous access tract unless >18 Fr.
d. reduces postoperative bleeding.
e. is associated with pain unrelated to tube diameter.

22. A small-caliber (8 to 18 Fr) compared with a large-caliber (20 to 24 Fr) nephrostomy tube after percutaneous renal surgery is associated with:
    a. equivalent pain.
    b. more urinary leakage.
    c. less postprocedure blood loss.
    d. less need for removal in the radiology suite.
    e. earlier hospital discharge.
23. Adjuncts intended to enhance hemostasis of the percutaneous tract include all EXCEPT:
    a. direct cauterization of the tract.
    b. microwave treatment of the tract.
    c. cryotreatment of the tract.
    d. insertion of oxidized cellulose.
    e. instillation of fibrin glue.
24. Compared with internal ureteral stents after percutaneous renal surgery, nephrostomy tubes are associated with:
    a. reduced need for a second procedure for removal.
    b. greater technical success rate.
    c. greater narcotic use.
    d. fewer complications.
    e. less urinary leakage from skin entry site.
25. Following an unremarkable percutaneous nephrolithotomy, there is nonpulsatile bleeding from the tract when the sheath is removed around a 12-Fr nephrostomy tube. The next step is:
    a. replace the nephrostomy tube with an 18-Fr Malecot catheter.
    b. replace the nephrostomy tube with a ureteral stent and suture the skin.
    c. irrigate the nephrostomy tube.

d. occlude the nephrostomy tube and apply pressure to the incision.
e. replace the nephrostomy tube with a Kaye nephrostomy tamponade balloon.

26. During a percutaneous resection of a 2-cm upper pole urothelial neoplasm, there is sudden hemorrhage from the resection site. The next step is:
    a. continue with the procedure if vision is adequate.
    b. insert a percutaneous nephroureteral stent.
    c. instill gelatin granules plus thrombin into the collecting system.
    d. place an 18-Fr Councill catheter with the balloon inflated at the injury site.
    e. prepare the patient for selective angioembolization.
27. A 65-year-old man calls the office 1 week after percutaneous nephrolithotomy complaining of bright red blood in the urine on his last two urinations. He is otherwise feeling well. He should next:
    a. check the percutaneous access site and come to the hospital if there is external bleeding.
    b. force fluids and call back if bleeding persists.
    c. take aminocaproic acid (Amicar).
    d. apply pressure to the percutaneous access site.
    e. come to the hospital.
28. Which of the following has NOT been reported to cause renal pelvic perforation in association with percutaneous renal surgery?
    a. Wire passage
    b. Tract dilation
    c. Massive hemorrhage
    d. Use of resectoscope
    e. Ultrasonic lithotripsy
29. Two days after percutaneous endopyelotomy in a 65-year-old woman, nephrostography reveals contrast entering the colon. The next step is to:
    a. perform exploratory laparotomy.

b. maintain the nephrostomy tube in place and insert a ureteral stent.
c. maintain the nephrostomy tube in place and insert a colostomy tube.
d. back out the nephrostomy tube into the colon and insert a new nephrostomy tube.
e. start parenteral feeding, after appropriate tube insertions.

30. Injury to which organ during percutaneous renal surgery can often be managed with little additional interventions?
    a. Liver
    b. Spleen
    c. Duodenum
    d. Jejunum
    e. Gallbladder
31. Regarding pleural injuries in association with percutaneous renal surgery:
    a. access below the 12th rib results in hydropneumothorax in 1% to 2% of cases.
    b. supra-12th rib punctures (the 11th intercostal space) result in hydropneumothorax in 20% to 40% of cases.
    c. supra-11th rib punctures (the 10th intercostal space) result in hydropneumothorax in 50% to 75% of cases.
    d. combined with distal ureteral obstruction, a nephropleural fistula can occur.
    e. thoracostomy to water seal drainage and suction is recommended.
32. Irrigation fluid during percutaneous renal surgery:
    a. is not absorbed systemically unless there is significant venous injury.
    b. should be normal saline except during percutaneous nephrolithotomy.
    c. can have fatal consequences.
    d. should not be glycine.
    e. will not create a defined extrarenal collection.
33. A 55-year-old woman has an oral temperature of 38.5°C on the first night after an uncomplicated percutaneous nephrolithotomy for a

partial staghorn renal calculus. A nephrostomy tube is in place. She is hemodynamically stable. The preoperative urine culture had grown a pansensitive *Proteus* sp., and she had received oral trimethoprim sulfamethoxazole for 2 weeks preoperatively. One gram of cefazolin had been administered on call to the operating room. The next step is:

a. observation.
b. culture aspirate from nephrostomy tube and irrigate nephrostomy tube.
c. Doppler ultrasonography of lower extremities and/or pulmonary embolus-protocol computed tomography scan.
d. administer broad-spectrum antibiotics.
e. culture urine and blood, obtain chest radiograph, and administer broad-spectrum antibiotics.

34. Following percutaneous renal surgery, loss of renal function is:
    a. approximately 5% of ipsilateral function per access site.
    b. minimal in the absence of vascular injury.
    c. greater than after shock wave lithotripsy.
    d. less in pelvic compared with orthotopic kidneys.
    e. greater in solitary compared with nonsolitary kidneys.
35. With regard to ureteral stents, which factors have been associated with improved patient tolerance?
    a. Ureteral stent length
    b. Ureteral stent diameter
    c. Stent material
    d. Distal stent tip architecture (pigtail vs. Polaris)
    e. None of the above

Answers

1. **d. Urinary retention.** Obstruction of the lower urinary tract is best treated by drainage of the bladder rather than the kidney, unless secondary obstruction of the upper tract has developed that is refractory to vesical drainage. The other indications are appropriate ones for percutaneous nephrostomy.

2. **b. Requires less anesthesia.** Percutaneous nephrostomy can be done under local anesthesia, as opposed to retrograde ureteral stent placement, which usually requires at least intravenous sedation, and commonly general or regional anesthesia. Percutaneous nephrostomy has a greater initial success rate than retrograde ureteral stent placement, at least when the collecting system is dilated. Percutaneous nephrostomy is commonly associated with bacteriuria and has health-related quality-of-life scores that are equivalent to those associated with retrograde ureteral stent placement. Ureteral stents provide satisfactory drainage in most cases of ureteral obstruction owing to malignancy.
3. **e. Immediately posterior to the kidneys are the quadratus lumborum muscle, the psoas muscle, and the diaphragm.** The upper poles are anterior to attachments of the diaphragm. It is the left kidney that is slightly cephalad to the right one. The second two statements are correct, except that the angulation is 30 degrees rather than 45 degrees. The apposition of the colon to the kidney varies with location; it is greatest on the left side but at the lower rather than upper pole.
4. **a. Paired anterior and posterior calyces enter the infundibula approximately 90 degrees from each other.** The paired anterior and posterior calyces enter approximately 90 degrees from each other. **Although compound calyces are common in the lower pole, they are almost always present in the upper pole.** In approximately two-thirds of kidneys, there are only two major calyceal systems (upper and lower). There are 5 to 14 minor calyces in each kidney. Because variation is considerable, the lateral-medial orientation of the calyces on anteroposterior radiography cannot be used to reliably determine which calyces are posterior.
5. **c. Segmental, interlobar (infundibular), arcuate, interlobular.**
6. **a. All patients should receive prophylactic antimicrobials.** The American Urological Association recommends periprocedural antimicrobial prophylaxis for all cases of percutaneous renal surgery. Urine cultures are considered standard only in patients where bacteriuria is likely; in other cases a screening urinalysis

likely is adequate, with urine culture when the urinalysis is suspicious. The urine cannot be sterilized in some patients, especially in the presence of an externalized urinary catheter or an infected calculus, and the goal in these situations is only to suppress the bacterial count before intervention. Aminoglycosides (e.g., gentamicin) are acceptable for antimicrobial prophylaxis when combined with another agent. Ampicillin/sulbactam, first- and second-generation cephalosporins, and fluoroquinolones are acceptable single agents for antimicrobial prophylaxis.

7. **d. Warfarin is 5 days. The recommended preoperative cessation periods are as follows: herbal medicines, 1 week; clopidogrel, 5 days; aspirin, 1 week; warfarin, 5 days; nonsteroidal inflammatory agents, 3 to 7 days.**
8. **d. Balloon dilation compared with semirigid plastic dilation of the access tract.** There is no evidence that balloon dilation is associated with less pain compared with semirigid plastic dilation of the access tract. All of the other maneuvers have been demonstrated in randomized controlled clinical trials to reduce pain associated with percutaneous renal access.
9. **e. Reduced pressure in the collecting system.** The angle of the sheath is more horizontal in the supine compared with the prone position for percutaneous renal surgery, which reduces pressure in the collecting system (the volume also is reduced, which is a disadvantage). When padding is appropriate, pulmonary mechanics are better in the prone position. The prone position also provides a large horizontal working surface and easier entry into posterior and upper pole calyces compared to the supine position.
10. **a. Upper pole posterior calyx.** This offers the most versatile access to the intrarenal collecting system, and as long as the entry is below the 11th rib, the advantages generally outweigh the risks. Percutaneous access into an infundibulum or the renal pelvis poses a greater risk of vascular injury than calyceal entry. Middle calyceal access provides good access to the ureter but usually does not provide good access to the upper and lower calyces. In the prone position an anterior calyx offers little access to the rest of the kidney.

11. **e. Retrograde placement of externalized (single pigtail) ureteral stent for drainage.** This can be performed at the conclusion of the procedure for drainage as an alternative to a nephrostomy tube, but it is not useful before access because the pigtail might interfere with the procedure and it would not have any advantage over a straight ureteral catheter or an occlusion balloon catheter. The other choices are all well-described techniques of retrograde assistance for percutaneous renal access.
12. **e. Is more traumatic.** A 21-gauge needle is not as easy to direct as an 18-gauge needle because it is more flexible. A 21-gauge needle requires a 0.018-inch guidewire, and because of this extra step (exchanging the 0.018-inch guidewire for a 0.035-inch guidewire) there is a greater risk of loss of access. Compared with an 18-gauge needle, the 21-gauge needle is less traumatic; this is its primary advantage, and it is for this reason that the 21-gauge needle should be used when the operator is less experienced or if minimizing trauma is paramount.
13. **d. Visualizes the access needle better.** It is easier to see a needle and monitor tract dilation with fluoroscopy than with ultrasonography. Percutaneous access is always more difficult in the morbidly obese, and ultrasonography is no better than fluoroscopy in this situation. Ultrasonography is more portable, can more rapidly evaluate different views of the kidney, and is preferred in settings in which retrograde access cannot be attained or is difficult to attain (kidneys above urinary diversions, transplanted kidneys, kidneys above a completely obstructed ureter, etc.).
14. **e. Continuously monitors depth of needle penetration.** The "triangulation" technique monitors depth of needle placement in all fluoroscopic views, whereas the "eye-of-the-needle" technique assesses depth only at the final step. If the fluoroscopy field is collimated down and the needle is held with a hemostat, sponge forceps, or purpose-built needle holder, then radiation exposure to the operator's hands can be avoided with both techniques. Retrograde assistance is useful with any fluoroscopic percutaneous

renal access, and both techniques are more difficult in morbidly obese patients.

15. **c. Least expensive with metal dilators.** Metal dilators are least expensive on a per-case basis because they are reusable. The metal dilators are also the most effective dilators. It is uncertain which are the safest dilators, at least in terms of association with hemorrhage. The balloon dilators are more effective in hypermobile kidneys than other techniques and also more rapid than sequential passage of metal or semirigid dilators. The one-shot semirigid dilator technique requires considerable manual force to create the tract.
16. **e. Lower hemorrhage rates than in normal kidneys can be expected.** The rate of major hemorrhagic complications during percutaneous renal surgery in horseshoe kidneys (4.3%) is less than in normal kidneys (6% to 20%). Upper pole access is useful, and direct lower pole access is usually not possible. Cross-sectional imaging is useful in assessing the anatomy of horseshoe kidneys. **The initial entry into a horseshoe kidney usually is more medial than in normal kidneys.**
17. **a. Retrograde assistance is difficult.** Because of the site of ureteral implantation in the bladder, retrograde assistance for percutaneous renal surgery in transplant kidneys is difficult. Given this, ultrasonography is more useful than fluoroscopy for guiding initial access. Semirigid plastic and metal dilators are often more useful than balloon dilators because of the perinephric scarring, which makes the transplant kidney quite fixed in place rather than hypermobile. Despite the challenges, percutaneous renal surgery in transplant kidneys has a high success rate, and secondary procedures are usually not required.
18. **d. Stay more securely in the kidney than Malecot catheters.** Malecot tubes are the easiest to pull out, and circle nephrostomy tubes are the hardest. All tubes should be secured at the skin to reduce the risk of at least one mechanism of tube removal. A ureteral catheter can be passed through the end hole of a Councill catheter; a Foley catheter does not have this end hole. Saline or water should be used to inflate the balloon, because the more

viscous contrast material might hinder emptying of the balloon when removal is attempted. All nephrostomy tubes, even ones with robust internal retention devices, should be fixed to the skin externally with a suture or other mechanism. There is no evidence that Foley and Malecot catheters differ in propensity for infection.

19. **a. Is used in nephroureteral stents.** The Cope retention mechanism is used in the renal pelvic portion of nephroureteral stents. According to one study, the Cope retention mechanism is more secure than Malecot wings but does not retain as well as a balloon. It is more secure than a passive pigtail retention mechanism owing to the string that holds the coil in place. There is no evidence that more than one Cope-type nephrostomy tube should not be used.
20. **a. Maintenance of the working sheath.** The stiff working access sheath would be a poor choice for postprocedure collecting system drainage. All of the other options have been described.
21. **a. Offers greater assurance of upper urinary tract drainage than an internal ureteral stent.** Drainage of upper urinary tract after percutaneous renal surgery is adequate with an internal ureteral stent in most cases (or with no tube at all in selected cases), but when hemorrhage occurs, the larger caliber of a nephrostomy tube provides better drainage of the upper urinary tract collecting system than an internal ureteral stent. The nephrostomy tube does not have to be placed in the dilated access site (i.e., it can be placed at a new site), although that is common practice. Although redilation may be required, any external nephrostomy tube maintains the percutaneous access tract. There is actually less hemorrhage when a postoperative nephrostomy tube is omitted. Most studies suggest that the pain associated with nephrostomy tubes is related to tube diameter, with smaller-caliber tubes causing less pain.
22. **d. Less need for removal in the radiology suite.** The removal of larger tubes occasionally can be followed by immediate hemorrhage; this is rare with smaller tubes. Therefore large-caliber nephrostomy tubes should be removed in a radiology suite where there is the opportunity for immediate replacement of the tube.

Small-caliber tubes can be removed safely at the bedside after a period of clamping to assess clinically for distal ureteral obstruction. A number of studies have compared the impact of nephrostomy tube diameter after percutaneous renal surgery. Only one study found no benefit to the smaller tube. Otherwise, consistent advantages of the small-caliber tubes were less pain, less urinary leakage, and no change in postprocedure blood loss. There is no consistent evidence that small-caliber tubes are associated with shorter duration of hospitalization compared with large-caliber tubes.

23. **b. Microwave treatment of the tract.** Microwave treatment of the tract would be difficult with current instruments. The other options have all been described. Other hemostatic agents that have been inserted/instilled into the tract include gelatin sponge and gelatin granules plus thrombin.
24. **c. Greater narcotic use.** Most randomized controlled trials comparing internal ureteral stents to large-caliber nephrostomy tubes after percutaneous renal surgery have shown reduced narcotic use in the stented patients. The difference is less significant when a small-caliber nephrostomy tube is used. Depending on physician preference, both internal ureteral stents (if attached to a string that exits via the flank) and small-caliber nephrostomy tubes can be removed at the bedside. Randomized controlled trials comparing internal ureteral stents to nephrostomy tubes have not revealed any difference in technical success rates, complication rates, or incidence of urinary leakage from the skin entry site.
25. **d. Occlude the nephrostomy tube and apply pressure to the incision.** The first step in this situation is to occlude the nephrostomy tube and apply pressure to the incision. Let the collecting system clot off, and do not irrigate until the following morning. This management is successful in the majority of cases. If bleeding persists, then insert a Kaye nephrostomy tamponade balloon. An 18-Fr Malecot catheter will be no more effective than the 12-Fr Cope nephrostomy tube. Irrigation is not useful, and removing the nephrostomy tube altogether is ill advised.

26. **a. Continue with the procedure if vision is adequate.** If the procedure can be continued with acceptable vision, then the blood loss cannot be great. However, if vision is lost, then the procedure must be aborted. If so, then inserting and occluding a nephrostomy tube, as well as applying pressure to the incision so that the collecting system clots off, will suffice in most cases. If this is not successful, then place a Councill catheter and attempt to inflate the balloon at the injury site. Instillation of gelatin granules plus thrombin into the collecting system can create a clot that is difficult to manage. Selective angioembolization is required only when an arterial injury does not respond to less intensive management, or if the injury is obviously a significant one that will not respond to these maneuvers.
27. **e. Come to the hospital.** Any report of bright red blood in the urine after percutaneous renal surgery should prompt hospital admission. This man likely has an arteriovenous fistula or arterial pseudoaneurysm. The conservative measures are not likely to be helpful, and aminocaproic acid (Amicar) is contraindicated in the setting of upper tract hemorrhage.
28. **c. Massive hemorrhage.** The renal pelvis will clot off before the pressure from hemorrhage would rupture it. Any manipulation during percutaneous renal surgery can cause renal pelvic perforation.
29. **d. Back out the nephrostomy tube into the colon and insert a new nephrostomy tube. The main principle of care of a colon injury associated with percutaneous renal surgery is prompt and separate drainage of the colon and urinary collecting system.** If detected postoperatively, the simplest management is to back the nephrostomy tube out of the kidney and into the colon to serve as a colostomy, and then obtain separate access to the upper urinary tract, with either a new percutaneous access that does not traverse the colon or a retrograde-placed ureteral stent. Parenteral feeding is usually not required, and for the typical extraperitoneal injury, open surgical repair usually is needed only if the patient develops peritonitis or sepsis.

30. **a. Liver.** Although splenic injuries have been managed conservatively, the need for surgical intervention is more likely than with liver injuries. Injuries to small bowel or the biliary system require prompt treatment.
31. **d. Combined with distal ureteral obstruction, a nephropleural fistula can occur.** Nephropleural fistula (urinothorax) is a direct and persistent communication between the intrarenal collecting system and the intrathoracic cavity, which can follow percutaneous renal access of the upper urinary tract in the setting of pleural transgression. Some degree of distal ureteral obstruction usually contributes to the problem. The rates of pleural injures for infra-12th rib, supra-12th rib, and supra-11th rib punctures are approximately less than 0.5%, 5%, and 25%, respectively. Thoracostomy is not necessary for all patients with hydrothorax. If one is needed, then a small-caliber tube with a Heimlich valve is all that is required in the absence of lung injury.
32. **c. Can have fatal consequences. Intravascular hemolysis from the extravasated water irrigant can be fatal. The irrigant for percutaneous renal surgery should be saline, with the exception of glycine or similar nonelectrolytic isotonic fluids when monopolar electrocautery is used.** Intravascular or extravascular extravasation of fluid from continued irrigation in the setting of a large venous injury or collecting system perforation can lead to clinically significant sequelae including volume overload and extrarenal fluid collections that require drainage.
33. **a. Observation.** Preoperative and perioperative management of this patient has been appropriate. In this setting, most patients with fever after percutaneous nephrolithotomy do not have infection. If the fever is an isolated postoperative one, then standard postoperative care (early ambulation, use of incentive spirometry, etc.) is all that is needed. If the fever does not resolve promptly, then appropriate diagnostic evaluation and initiation of antimicrobial therapy and other supportive care are indicated.
34. **b. Minimal in the absence of vascular injury.** The kidney suffers little permanent damage after uncomplicated percutaneous renal surgery. If there is significant loss of function, it is usually owing to

disastrous vascular injury or the angioembolization used to treat hemorrhage. Loss of renal function associated with percutaneous renal surgery is less than or equal to the loss associated with shock wave lithotripsy. There is no evidence that damage to the kidney is any more or less in pelvic, orthotopic, solitary, or nonsolitary kidneys.

35. **e. None of the above.** Randomized controlled trials have failed to show association between stent length, diameter, material, or architecture and impact on patient tolerability.

Chapter Review

1. Percutaneous nephrostomy and retrograde ureteral stents are generally equivalent in their capacity to resolve fever in patients with upper urinary tract obstruction. However, obstruction complicated by infection is an emergency, and in the unstable patient, percutaneous drainage may be more efficacious.
2. The colon can be lateral or posterior to the right and left kidney.
3. A guidewire that enters the kidney percutaneously and exits the urethra via the meatus (through and through access) may be the only guidewire used when operating on the upper urinary tract. However, in all other situations, two guidewires—a safety and a working guidewire—are required. No matter what the access, it is always prudent to have a safety guidewire in addition to the working guidewire.
4. It is imperative that the dilators do not pass too far into the collecting system because this results in renal pelvic injury.
5. Percutaneous nephrostomy is generally the preferred approach for endoscopy of the obstructed collecting system in the transplanted kidney.
6. Approximately 1% of percutaneous procedures are complicated by delayed hemorrhage. Delayed hemorrhage is usually due to arteriovenous fistulas or arterial pseudoaneurysms. The preferred management is selective angioembolization.

7. Renal arteries are end arteries and result in loss of the segment of renal parenchyma they supply when occluded. Renal veins communicate with each other.
8. Complications of percutaneous nephrostomy include hemorrhage, collecting system injury, colon injury, pleural injury, neuromuscular injuries, air embolism, and infundibular stenosis.
9. Although compound calyces are common in the lower pole, they are almost always present in the upper pole.
10. The recommended preoperative cessation periods are as follows: herbal medicines, 1 week; clopidogrel, 5 days; aspirin, 1 week; warfarin, 5 days; nonsteroidal inflammatory agents, 3 to 7 days.
11. The main principle of care of a colon injury associated with percutaneous renal surgery is prompt and separate drainage of the colon and urinary collecting system. The simplest management is to back the nephrostomy tube out of the kidney and into the colon to serve as a colostomy, and then obtain separate access to the upper urinary tract, either with a new percutaneous access that does not traverse the colon or with a retrograde-placed ureteral stent.
12. Intravascular hemolysis from the extravasated water irrigant can be fatal. The irrigant for percutaneous renal surgery should be saline, with the exception of glycine or similar nonelectrolytic isotonic fluids when monopolar electrocautery is used.

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Principles of Urologic Endoscopy

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Keywords

cystoscopy; endoscopy; equipment; guidelines; history; technique; ureteroscopy

Questions

1. Patients undergoing diagnostic cystoscopy should receive prophylactic antibiotics if they have any of the following risk factors except:
    a. poor nutritional status.
    b. anatomic anomalies.
    c. hypertension.
    d. corticosteroid use.
    e. smoking history.
2. Compared with reusable flexible ureteroscopes, single-use flexible ureteroscopes have been shown to have:
    a. decreased active deflection.
    B. comparable visibility.
    c. well-established cost savings.
    d. decreased durability.
    e. longer procedure time.
3. Techniques that have been shown to improve flexible cystourethroscopy tolerance in men include all of the following

except:

a. allowing the patient to observe the procedure.
b. having the patient empty his bladder before the procedure.
c. playing classical music during the procedure.
d. using lidocaine lubricating gel.
e. increasing the hydrostatic pressure of the irrigant during scope passage.

4. Indications for ureteroscopy include all of the following except:
    a. obstructing ureteral calculus.
    b. filling defect of the renal pelvis.
    c. 1.2-cm renal calculus in the lower pole.
    d. ureteropelvic junction obstruction with a large crossing vessel present.
    e. 1.5-cm midureteral stricture.
5. Which of the following statements about ureteroscopy are true?
    a. Ureteral access sheaths decrease intrarenal pressure during ureteroscopy.
    b. Flexible ureteroscopes accept working instruments 3.6 Fr in diameter.
    c. Normal saline should be used for irrigation during ureteroscopy.
    d. A preoperative antibiotic is needed only in high-risk patients.
    e. a and c.
6. Compared with white-light endoscopy, narrow-band imaging has been shown to:
    a. predict favorable response to bacillus Calmette-Guérin (BCG) intravesical therapy.
    b. accurately differentiate between low- and high-grade lesions.
    c. obviate the need for re-resection in patients with high-grade tumors involving the lamina propria.
    d. significantly improve detection accuracy of muscle-invasive lesions.
    e. improve detection of noninvasive lesions, including carcinoma in situ.
7. Which of the following statements are true?
    a. The holmium laser is absorbed in 3 cm of water.

b. Water is the preferred irrigant for ureteroscopy because of improved visibility.
c. Baskets made of nitinol are more “kink resistant” compared with stainless steel.
d. Balloon dilation of the intramural ureter is usually necessary before flexible ureteroscopy.
e. Compared with fiberoptic flexible ureteroscopes, digital flexible ureteroscopes less frequently require the use ureteral access sheaths.

8. Techniques to minimize staff radiation exposure include all of the following except:
   a. using “last image hold” setting.
   b. using fixed fluoroscopy units.
   c. surgeon control of the foot pedal.
   d. using image collimation.
   e. using pulse fluoroscopy mode.

Answers

1. **c. Hypertension.** The American Urological Association’s Best Practice Policy Statement on Antimicrobial Prophylaxis did **not recommend routine antibiotic administration for diagnostic cystourethroscopy.** This recommendation was based on the recognition that some randomized studies have shown antibiotic prophylaxis reduces bacteriuria and symptomatic infection rates, whereas others have not. However, **antibiotic administration was advocated** by the panel for patients with host factors increasing their risk of infection. These include **advanced age, anatomic anomalies, poor nutritional status, smoking, chronic corticosteroid use, immunodeficiency, chronic indwelling hardware infected with endogenous or exogenous material, distant coinfection, and prolonged hospitalization.**
2. **b. Comparable visibility.** One of the early studies evaluating disposable ureteroscopes was a cadaveric study comparing the LithoVue to nondisposable fiberoptic and digital ureteroscopes. The

researchers found the LithoVue to be comparable in visibility and scope maneuverability.

3. **b. Having the patient empty his bladder before the procedure.** A variety of prospective studies have been performed with the aim of improving patient tolerance during office-based diagnostic flexible cystourethroscopy. A meta-analysis of four randomized trials involving 411 patients found that patients receiving lidocaine gel were 1.7 times less likely to experience moderate to severe pain during the procedure. Results of a randomized trial involving 151 men indicated that increasing the hydrostatic pressure of the irrigation solution during passage of the scope through the membranous urethra was associated with significantly less discomfort on an analog pain scale. In another study, men who were allowed to watch the procedure had significantly less pain on a 100-mm visual analog scale. Last, 70 men were randomly assigned to hear either no music or classical music during cystourethroscopy. Patients listening to classical music had significantly less pain, greater satisfaction, lower postprocedure pulse rates, and lower systolic blood pressures.
4. **d. Ureteropelvic junction obstruction with a large crossing vessel present.** All of the options are common indications for ureteroscopy except ureteropelvic junction obstruction due to a large crossing vessel. Although ureteropelvic junction obstruction can be managed with ureteroscopic endopyelotomy, **patients with a large crossing vessel are better treated by laparoscopic pyeloplasty.**
5. **e. a and c.** Auge and colleagues measured the pressure within the renal pelvis and proximal, mid, and distal ureter before and after ureteral access sheath placement in five patients who had previously undergone nephrostomy tube placement. The pressure within the collecting system was found to be significantly lower at each location following access sheath placement.
6. **e. Improve detection of noninvasive lesions, including carcinoma in situ.** Narrow-band imaging uses only blue (415 nm) and green (540 nm) wavelengths to image the urothelium compared with white-light endoscopy, which uses the entire visible light

spectrum. Blue and green wavelengths are strongly absorbed by hemoglobin, improving visibility of urothelial capillaries, small papillary lesions, and carcinoma in situ. A meta-analysis of eight studies including 1022 patients found that **narrow-band imaging improves detection accuracy of noninvasive lesions, including carcinoma in situ.**

7. **c. Baskets made of nitinol are more "kink resistant" compared with stainless steel.** Nitinol has a variety of advantageous properties compared with stainless steel. Nitinol is more biocompatible, has greater torquability and improved "memory," and is more resistant to kinking. These properties make it ideal for stone basket construction. Holmium laser energy is absorbed by 3 mm of water. Saline is the preferred irrigation solution because of the decreased risk of "transurethral resection syndrome" compared with hypotonic solutions. The need for ureteral dilation has decreased over time with the advent of smaller-diameter ureteroscopes. However, digital models have larger tip and shaft diameters, making them more likely to need ureteral dilation or access sheath placement compared with their fiberoptic counterparts.
8. **b. Using fixed fluoroscopy units.** All endourologic procedures using fluoroscopy should operate on the ALARA principle (as low as reasonably achievable). Techniques that have been shown to minimize radiation exposure include surgeon control of the foot pedal, using "last image hold," image collimation, and pulsed fluoroscopy mode. Compared with fixed units, mobile C-arm fluoroscopy machines are able to position the image intensifier closer to the patient, thereby reducing exposure while improving image quality.

Chapter Review

1. Rigid cystourethroscopes are manufactured in sets consisting of an optical lens, bridge, sheath, and obturator.

2. Flexible cystourethroscopes are available in fiberoptic, standard-definition and high-definition digital models.
3. Cystourethroscope sizes are expressed using the French gauge system, with 1 Fr equaling a third of a millimeter in circumference.
4. The smallest-diameter cystourethroscope that can be used to perform the procedure should be selected to decrease the risk of genitourinary tract trauma.
5. Antimicrobial prophylaxis is not recommended for diagnostic cystourethroscopy unless patient-related risk factors are present.
6. Increasing the hydrostatic pressure during scope passage and allowing men to observe the procedure have been prospectively shown to improve patient comfort during flexible cystourethroscopy.
7. Semirigid ureteroscopy is used below the iliac vessels and flexible ureteroscopy above.
8. Semirigid ureteroscopes with two working channels permit better irrigation and the added safety of being able to pass a lithotripsy device through one channel when needing to fragment a stone engaged in a basket in the other channel.
9. Flexible ureteroscopes must be straight when passing a laser fiber or the working channel will be damaged.
10. The golden rule of safe laser lithotripsy: “Do not step on the pedal if you cannot see the tip of the fiber in contact with the stone.”
11. Stone baskets made of nitinol maintain their shape, resist kinking, and allow disengagement of stones more reliably that stainless steel baskets.
12. Ureteral access sheaths facilitate repeated passage of flexible ureteroscopes and decrease the intrapelvic pressure during ureteroscopy.
13. Mobile C-arm fluoroscopy is preferred because of greater mobility, improved image quality, and less scatter radiation exposure to the surgeon compared with urology tables with fixed fluoroscopy units.
14. A routine preoperative antibiotic should be given to all patients undergoing ureteroscopy.
15. Normal saline should be used as irrigation during ureteroscopy to prevent absorption of a hypotonic solution.

14

Fundamentals of Laparoscopic and Robotic Urologic Surgery

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Keywords

education; laparoscopic and robotic instrumentation; laparoscopy; minimally invasive surgery; robotic surgery

Questions

1. Absolute contraindications to laparoscopic surgery include all of the following EXCEPT:
    a. uncorrectable coagulopathy.
    b. hemodynamic instability.
    c. significant abdominal wall infection.
    d. suspected malignant ascites.
    e. extensive prior abdominal or pelvic surgery.
2. Of the following, which is considered a relative contraindication to laparoscopic surgery?
    a. Generalized peritonitis
    b. Massive hemoperitoneum
    c. Intestinal obstruction with intention to treat
    d. Extensive prior abdominal or pelvic surgery
    e. Abdominal wall infection
3. The most effective preoperative preparation for patients undergoing routine laparoscopic renal surgery is:

a. a 3-day mechanical bowel preparation if an extraperitoneal or retroperitoneoscopic approach is anticipated.
b. a mechanical bowel preparation and antibiotic preparation with neomycin and metronidazole.
c. for most uncomplicated patients, a clear liquid diet and a light mechanical bowel preparation the day before surgery.
d. both an antibiotic and 3-day mechanical bowel preparation in patients who have had previous abdominal surgery if one anticipates encountering dense intra-abdominal adhesions.
e. intravenous antibiotics 1 hour before surgery.

4. Which of the following statements regarding enhanced recovery after surgery (ERAS) protocol is TRUE?
   a. They reduce hospital stay by 30% but have no effect on complication rates.
   b. They reduce hospital stay by 30% and decrease complication rates by 50%.
   c. They decrease admission rates and have no effect on hospital stay.
   d. They increased healthcare costs.
   e. They have no effect on complication rates and decrease healthcare costs.
5. Which of the following is TRUE regarding operating table padding?
   a. The unaugmented operating room table mattress is best for reducing pressure.
   b. Shoulder support and braces should always be used.
   c. Gel padding is better than egg crate padding.
   d. Kidney rest should always be used for more than 30 minutes.
   e. Egg crate padding is equivalent to gel padding and both are better than the unaugmented bed.
6. Which of the following statements regarding pneumoperitoneum insufflation is TRUE?
   a. CO_2 as an insufflant can be dangerous because it can support combustion.
   b. CO_2 is most commonly used because it is insoluble in the blood.

c. In patients with chronic respiratory disease, CO_2 is advantageous because it does not accumulate in the bloodstream.
d. Argon gas would be an ideal insufflant because of its low cost and poor solubility in blood.
e. Nitrous oxide has previously been used for insufflation; however, it is no longer routinely used because of the potential for intra-abdominal explosion.

7. When a patient undergoing renal laparoscopy with a potential for extensive adhesions due to multiple prior abdominal surgeries, which of the following access techniques is recommended for obtaining a pneumoperitoneum and access to the abdomen for laparoscopy?
   a. Closed technique with Veress needle
   b. Closed technique with blind trocar insertion
   c. Open-access technique
   d. Hand-port access
   e. EndoTip entry
8. Of the following access locations, which port site most often requires formal closure with a fascial and peritoneal suture?
   a. 5-mm nonbladed ports
   b. 5-mm bladed ports
   c. 10- to 12-mm bladed ports placed on the midclavicular line
   d. 10- to 12-mm nonbladed ports placed on the midclavicular line
   e. 10- to 12-mm nonbladed ports placed on the anterior axillary line
9. Which of the following pneumoperitoneum pressures is associated with the least perturbation in cardiac parameters (i.e., change in stroke volume)?
   a. 12 mm Hg
   b. 15 mm Hg
   c. 18 mm Hg
   d. 21 mm Hg
   e. 24 mm Hg

10. Which of the following is a common physiologic effect that has been observed after the establishment of pneumoperitoneum?
    a. Increase in diaphragmatic motion
    b. Increase in disturbances of gastrointestinal motility
    c. Alkalosis
    d. Decrease in urinary output
    e. Increase in mesenteric vessel blood flow
11. Which of the following intra-abdominal structures is most commonly the site of injury in laparoscopic surgery?
    a. Bowel injury
    b. Vascular injury
    c. Liver injury
    d. Splenic laceration
    e. Bladder injury
12. What is a characteristic of a blunt trocar, compared with a bladed trocar?
    a. The blunt trocar requires formal closure of the port site regardless of its size.
    b. The blunt trocar takes less force to insert than the bladed trocar.
    c. The blunt trocar decreases the chance of injury to the epigastric vessels.
    d. The blunt trocar should only be placed in the midline.
    e. The blunt trocar eliminates possible trocar injury to the bowel.
13. All of the following are treatment options for a gas embolism during laparoscopy are true EXCEPT:
    a. Hyperventilate the patient with 100% oxygen.
    b. Immediately cease insufflation.
    c. Place the patient in a head-down position.
    d. Advance a central venous line into the right side of the heart.
    e. Place the patient in a right lateral decubitus position with the left side up.
14. Pneumomediastinum, pneumothorax, and pneumopericardium associated with laparoscopy are a result of:
    a. gas leaking along major blood vessels through congenital defects in the diaphragm.

b. gas passing through secondary enlargement of openings in the diaphragm.
c. diffusion of gas across the peritoneum and diaphragm.
d. a and b.
e. a and c.

15. If the Veress needle has been unintentionally placed in the iliac artery during creation of the pneumoperitoneum for insufflation of the abdomen, which of the following is the best course of action to minimize further patient injury?
    a. Remove the Veress needle, and proceed to open the abdomen.
    b. Remove the Veress needle, and then proceed with insufflating at a different location.
    c. Leave the Veress needle in place, and open the abdomen.
    d. Leave the Veress needle in place, and proceed with insufflation of the abdomen at a different location.
    e. Call for a vascular surgery consult.
16. What is the most effective management option if trocar injury to the iliac artery should occur during the placement of the first trocar?
    a. Remove the trocar, and open the abdomen immediately.
    b. Remove the trocar immediately, and proceed with reinsufflation of the abdomen and placement of the trocar at an alternate site.
    c. Leave the trocar in place, consult a vascular surgeon, and convert to open laparotomy.
    d. Leave the trocar in place, and proceed with insufflation of the abdomen and placement of another port at an alternate site.
    e. Remove the obturator, and immediately flush the port with fibrin glue.
17. Laparoscopic surgery can potentially cause thermal bowel injury during as a direct result from all of the following EXCEPT:
    a. capacitive coupling.
    b. insulation failure.
    c. inappropriate direct activation.
    d. electrode resistance.
    e. coupling to another instrument.

18. When a postoperative bladder injury is diagnosed following laparoscopy, what is the most effective method of treatment?
    a. Transurethral indwelling Foley catheter if it is an intraperitoneal injury of the bladder
    b. Open repair if it is an extraperitoneal injury of the bladder
    c. Laparoscopic or open repair if it is an intraperitoneal injury to the bladder
    d. Laparoscopic repair if it is an extraperitoneal injury to the bladder
    e. Transurethral injection of fibrin glue into the bladder injury site if it is an extraperitoneal injury to the bladder
19. Hypercarbia during laparoscopy may be related to all of the following EXCEPT:
    a. severe chronic respiratory disease.
    b. subcutaneous emphysema.
    c. increased insufflation pressures.
    d. prolonged operative time.
    e. radical nephrectomy.
20. Possible advantages of retroperitoneal laparoscopy include all of the following EXCEPT:
    a. less need for lysis of adhesions.
    b. decreased risk of paralytic ileus.
    c. decreased risk of port-site hernias.
    d. direct rapid access to the renal hilum.
    e. technically easier to learn.
21. In comparison with transperitoneal pelvic node dissection, which of the following incidence rates are higher after extraperitoneal pelvic lymph node dissection?
    a. Urinoma
    b. Lymphocele
    c. Bowel injury
    d. Laparoscopic repair if it is an extraperitoneal injury to the bladder
    e. Shoulder/hip pain
22. All of the following instruments might be part of a hemorrhage control tray EXCEPT:

a. laparoscopic needle drivers.
b. laparoscopic Satinsky clamp and accompanying trocar.
c. Lapra-Ty clip applier and 6-inch length of 3-0 absorbable suture.
d. hemostatic agents (fibrin glue, gelatin matrix thrombin, etc.) plus laparoscopic applicators.
e. laparoscopic renal biopsy forceps.

23. Which of the following hemostatic agents requires a 20-minute setup time before use?
    a. Tisseel
    b. FloSeal
    c. CrossSeal
    d. BioGlue
    e. CoSeal
24. Which of the following relationships is true for port placement for laparoscopic suturing?
    a. The angle produced by the horizontal plane and the instruments should be greater than 55 degrees, and the angle between the needle drivers should be less than 25 degrees.
    b. The angle produced by the horizontal plane and the instruments should be less than 55 degrees, and the angle between the needle drivers should be between 25 and 45 degrees.
    c. The angle produced by the horizontal plane and the instruments should be greater than 55 degrees, and the angle between the needle drivers should be greater than 45 degrees.
    d. The angle produced by the horizontal plane and the instruments should be less than 55 degrees, and the angle between the needle drivers should be less than 25 degrees.
    e. The angle produced by the horizontal plane and the instruments should be greater than 55 degrees, and the angle between the needle drivers should be between 25 and 45 degrees.
25. During a procedure using the da Vinci Robotic System, the robot malfunctions and one of the grasping forceps is closed on a vital structure. The system is completely unresponsive. The appropriate

action to safely disengage the instrument from the vital structure is to:

a. use the surgeon's console to override the system and robotically disengage the grasper.
b. remove the robotic instrument from the robotic arm.
c. use the sterile Allen wrench provided by the company to manually disengage the instrument and then remove it from the robotic arm.
d. use a handheld laparoscopic instrument to pry open the jaws of the robotic instrument.
e. unplug the surgeon's console and robotic tower, plug them back in, and restart the system.

26. Insufflation should never commence after placement of the Veress needle unless all of the following signs for proper peritoneal entry are confirmed EXCEPT?
    a. negative aspiration.
    b. easy irrigation of saline.
    c. negative pressure test.
    d. positive drop test.
    e. normal advancement test.
27. Insufflation of the abdomen is most commonly conducted using carbon dioxide because it is:
    a. noncombustible.
    b. rapidly absorbed.
    c. inexpensive.
    d. colorless.
    e. all of the above.
28. Helium is a useful insufflant in patients with:
    a. coronary artery disease.
    b. peripheral vascular disease.
    c. pulmonary disease.
    d. inflammatory bowel disease.
    e. chronic kidney disease.
29. Which of the following are signs of bowel insufflation with the Veress needle?
    a. Asymmetric abdominal distention

b. Flatus
c. High pressures reached after a large amount of CO_2 is insufflated
d. a and c
e. a and b

30. The diagnosis of air embolism is usually made by the anesthesiologist based on an initial abrupt:
    a. increase in end-tidal CO_2.
    b. decrease in end-tidal CO_2.
    c. increase in oxygen saturation.
    d. increase in mean arterial pressure.
    e. decrease in airway pressures.
31. Laparoscopic virtual reality (VR) trainers have been shown to:
    a. increase the operating time and improve the operative performance of surgical trainees with limited laparoscopic experience when compared with no training or with box-trainer training.
    b. decrease the operating time and improve the operative performance of surgical trainees with limited laparoscopic experience when compared with no training or with box-trainer training.
    c. decrease the operating time and improve the operative performance of surgical trainees with extensive laparoscopic experience when compared with no training or with box-trainer training.
    d. a and b.
    e. a and c.
32. When the patient is positioned in the modified flank position, the risk of developing rhabdomyolysis from flank pressure is increased by all of the following EXCEPT:
    a. body mass index (BMI) greater than 25.
    b. elevation of the kidney rest.
    c. age less than 45 years.
    d. male gender.
    e. full table flexion.

33. When using a laparoscopic stapling device, the 2.0-mm or 2.5-mm staple cartridges are preferred for:
    a. bowel.
    b. bladder.
    c. ureter.
    d. vascular (renal artery or vein).
    e. a and d.
34. All of the following represent options for port site fascial closure EXCEPT:
    a. retractors and direct vision.
    b. Endo Stitch.
    c. Carter-Thomason needlepoint suture passer.
    d. disposable Endo Close suture carrier.
    e. angiocatheter technique.
35. The basic principles of Hem-o-Lok clip placement include all of the following EXCEPT:
    a. incomplete circumferential dissection of the vessel.
    b. visualization of the curved tip of the clip around and beyond the vessel.
    c. confirmation of the tactile snap when the clip engages.
    d. during transaction of vessels, only a partial division is performed initially to confirm hemostasis before complete transaction.
    e. no cross clipping.
36. Balloon trocars are advantageous because they can help reduce the risk of:
    a. air embolism.
    b. alkalosis.
    c. subcutaneous emphysema.
    d. hypothermia.
    e. all of the above.
37. To avoid local or distant transmitted thermal injury during monopolar electrosurgery, certain precautions must be followed, including:
    a. checking the insulation of the electrosurgical instrument carefully for damage.

b. not activating the electrosurgical probe unless the metal part is in complete view.
c. not activating the probe unless it is in direct contact with the tissue to be incised.
d. never using a metal trocar in conjunction with an outer plastic retaining ring.
e. all of the above.

Answers

1. **e. Extensive prior abdominal or pelvic surgery.** Absolute contraindications for laparoscopic surgery include uncorrectable coagulopathy, intestinal obstruction, abdominal wall infection, massive hemoperitoneum or hemoretroperitoneum, generalized peritonitis or retroperitoneal abscess, and suspected malignant ascites.
2. **d. Extensive prior abdominal or pelvic surgery.** When extensive intra-abdominal or pelvic adhesions are suspected, close attention must be given to access into the abdomen, whether this is by Veress needle (Ethicon Endo-Surgery, Blue Ash, OH) or some open-access technique. The Palmer point (subcostal midclavicular line) on the left side is preferred. Alternatively, in these patients, a retroperitoneal approach may be preferable to a transperitoneal approach, but this is only a relative contraindication to performing laparoscopic surgery. All of the other options listed are absolute contraindications to laparoscopic surgery.
3. **e. Intravenous antibiotics 1 hour before surgery.** For extraperitoneoscopy and retroperitoneoscopy, no bowel preparation is necessary. Similarly, for transperitoneal laparoscopic/robotic procedures not involving the use of bowel segments for urinary tract reconstruction, a mechanical bowel preparation is not necessary. A recent large-scale propensity score-matched analysis demonstrated no benefit for mechanical bowel preparation in operative time, postoperative stay, or overall complications for patients undergoing laparoscopic nephrectomy.

4. **b. They reduce hospital stay by 30% and decrease complication rates by 50%.** These protocols have resulted in shorter hospital stay by 30%, a reduction in complications by 50%, decrease in readmission rates, and thus reducing healthcare costs.
5. **a. The unaugmented operating room table mattress is best for reducing pressure.** Use of the kidney rest was believed to be the most detrimental, and its use beyond 20 to 30 minutes was discouraged. Male patients with a BMI of 25 or higher undergoing laparoscopic surgery in the lateral position with the kidney rest elevated and the table completely flexed are at highest risk of developing rhabdomyolysis from flank pressure. In this study the unaugmented operating table mattress was superior to egg crate or gel padding as an augmenting surface material; of note, egg crate padding was equal or superior to the more expensive gel padding. Support or braces should never be used in Trendelenburg position, due to risk of brachial plexus injury.
6. **e. Nitrous oxide has previously been used for insufflation however, it is no longer routinely used because of the potential for intra-abdominal**; explosion. Most commonly, CO_2 is used as the insufflant because it does not support combustion and is very soluble in blood. However, in patients with chronic respiratory disease, CO_2 may accumulate in the bloodstream to dangerous levels. In these patients, helium may be used for insufflation once the initial pneumoperitoneum has been established with CO_2. The drawback of helium is that, like air, it is much less soluble in the blood than is CO_2. However, its use averts problems with hypercarbia. Other gases that were once used for insufflation, including room air, oxygen, and nitrous oxide, are no longer routinely used because of their potential side effects, such as air embolus or intra-abdominal explosion and potential to support combustion.
7. **c. Open-access technique.** Pneumoperitoneum can be more easily and, in one's early experience, more safely established using an open technique, especially in patients with multiple prior surgeries, who are at high risk for intra-abdominal adhesions. However, its

use involves making a larger incision and increases the chances of port-site gas leakage during the procedure. Studies in general surgery have shown the open technique to be as efficient as a closed approach. Insertion should be performed in an unscarred quadrant of the abdomen. If there is no scar-free area, then an open technique should be used.

8. **c. 10- to 12-mm bladed ports placed on the midclavicular line.** All bladed port sites that are greater than 5 mm should be formally closed, independent of location.
9. **a. 12 mm Hg.** Recent studies support a pneumoperitoneum pressure of 12 mm Hg, because this results in no perturbation in cardiac parameters (i.e., no change in stroke volume) versus a pressure of 15 mm Hg. Working at lower pneumoperitoneum pressures has also been found to reduce postoperative pain. In addition, a marked reduction in oliguria has been associated with working at 10 mm Hg pressure.
10. **d. Decrease in urinary output.** Because of increased intra-abdominal pressure from the pneumoperitoneum, diaphragmatic motion is limited. Laparoscopic surgery causes less significant disturbances of the gastrointestinal motility pattern compared with open surgery. Insufflation with CO_2 results in variable amounts of gas absorption, thereby raising the PCO_2 in the blood and creating an acidosis. Increased intra-abdominal pressure was found to be associated with a significant decrease in urinary output secondary to decreased blood flow to the renal cortex with an associated decrease in renal vein blood flow of up to 90% at 15 mm Hg.
11. **b. Vascular injury.** The most common site of injury during laparoscopic surgery, in reports in the literature, is vascular in origin, occurring in 2.8% of patients, followed by bowel injury at 1.1%. The most often injured intra-abdominal organ was the bowel, at an incidence of 1.2%.
12. **c. The blunt trocar decreases the chance of injury to the epigastric vessels.** The use of only blunt trocars decreases the chance of injury to the epigastric vessels by fivefold. Indeed, studies have shown the risk of inferior epigastric injury or port site

herniation is fivefold less with blunt versus sharp trocars. In addition, a recent meta-analysis demonstrated a lower relative risk of trocar site bleeding (3% vs. 9%) and overall complications (3% vs. 10%) with blunt compared with bladed trocars.

13. **e. Place the patient in a right lateral decubitus position with the left side up.** The treatment for a suspected gas embolism is immediate cessation of insufflation and prompt desufflation of the peritoneal cavity. The patient is turned into a **left lateral decubitus** head-down position (i.e., right side up) to minimize right ventricular outflow problems. The patient is hyperventilated with 100% oxygen. Advancement of a central venous line into the right side of the heart with subsequent attempts to aspirate the gas may rarely be helpful. Use of hyperbaric oxygen and cardiopulmonary bypass have also been reported.
14. **d. a and b.** Gas leaking along major blood vessels through congenital defects or secondary enlargement of openings in the diaphragm may lead to pneumomediastinum, pneumopericardium, and pneumothorax.
15. **d. Leave the Veress needle in place, and proceed with insufflation of the abdomen at a different location.** If vascular injury should occur with the Veress needle, the needle should be left in place to identify the area of injury, and insufflation of the abdomen can be reperformed at an alternate site and then the laparoscope inserted to identify the area of injury and to observe this as the Veress needle is removed to control any hemorrhage that may occur from the site. Note this is different than a trocar vessel injury.
16. **c. Leave the trocar in place, consult a vascular surgeon, and convert to open laparotomy.** A trocar injury to a major arterial vessel is a potentially life-threatening complication. The trocar should remain in place to tamponade the bleeding and also identify the area of injury once the abdomen is opened. The patient's blood should be typed and crossmatched, and immediate laparotomy should be performed and the site of vascular injury identified. A vascular surgery consult may be needed.

17. **d. Electrode resistance.** Electrosurgically induced thermal injury may occur through of one of four mechanisms: inappropriate direct activation; coupling to another instrument; capacitive coupling; and insulation failure.
18. **c. Laparoscopic or open repair if it is an intraperitoneal injury to the bladder.** When bladder injury is diagnosed postoperatively, the surgeon must determine whether the perforation is extraperitoneal or intraperitoneal. Extraperitoneal injury, without any complicating additional problems, may be treated by simple placement of a transurethral indwelling Foley catheter. Intraperitoneal injury is an indication for subsequent laparoscopic or open repair.
19. **e. Radical nephrectomy.** The potential for developing hypercarbia exists during both transperitoneal and preperitoneal laparoscopic procedures. Conceivably, this assumes greater importance in patients with preexisting airway and cardiovascular compliance. Vigilant perioperative anesthetic management is essential to prevent the development of potential complications related to CO_2 buildup. A rise in end-tidal CO_2 should prompt the anesthesiologist to adjust the respiratory rate and tidal volume to enhance CO_2 elimination. Simultaneously, the insufflation pressure of CO_2 should be decreased by the surgeon or, if need be, the operation should be halted and the abdomen desufflated until the end-tidal CO_2 returns to an acceptable level.
20. **e. Technically easier to learn.** Retroperitoneoscopy is associated with unique anatomic orientation and a relatively restricted initial working area compared with transperitoneal laparoscopy. This results in a steeper learning curve.
21. **b. Lymphocele.** Absence of the peritoneal absorptive surface after extraperitoneoscopic lymphadenectomy may increase the risk of development of postoperative lymphocele
22. **e. Laparoscopic renal biopsy forceps.** The contents of a hemorrhage tray for laparoscopic surgery include the following:
    - Laparoscopic Satinsky clamp (Medline Industries Inc., Mundelein, IL)

- 10-mm suction/irrigation tip
- Endo Stitch device with a 4-0 absorbable suture
- Lapra-Ty clip (Ethicon US, LLC, CA) applier and a packet of Lapra-Ty clips
- 6-inch length of 4-0 vascular suture on an SH needle with a Lapra-Ty clip preplaced on the end
- Two laparoscopic needle drivers
- Topical hemostatic agent of choice

23. **a. Tisseel.** Tisseel (Baxter, Glendale, CA) is a form of fibrin glue containing fibrinogen, calcium chloride, aprotinin, and thrombin. It is useful as a topical hemostatic agent as well as a tissue glue, but it has a 20-minute setup time and thus must be prepared well in advance of potential use.
24. **b. The angle produced by the horizontal plane and the instruments should be less than 55 degrees, and the angle between the needle drivers should be between 25 and 45 degrees.** Frede and colleagues performed an in vitro experiment performing laparoscopic suturing while varying trocar relationship to the horizontal plane and the distance between the two instrument trocars. They found that suturing was easiest when the angle between the horizontal plane and the instruments was less than 55 degrees and the angle between the two instruments was between 25 and 45 degrees.
25. **c. Use the sterile Allen wrench provided by the company to manually disengage the instrument and then remove it from the robotic arm.** In the event of a system failure of the da Vinci Robotic System (Intuitive Surgical, Sunnyvale, CA) during which the robotic arms are rendered nonfunctional, instrument jaws can be manually opened using a sterile Allen wrench provided by the company for this purpose.
26. **c. Negative pressure test.** Several tests can be performed in an attempt to confirm proper placement of the Veress needle within the peritoneal cavity before insufflation, to reduce the risk of insufflation-related complications. These tests include: the aspiration/irrigation/aspiration test, the advancement test, and the drop test. Insufflation should never be initiated unless all of the

signs for proper peritoneal entry (negative aspiration, easy irrigation of saline, negative aspiration of saline, positive drop test, and normal advancement test) have been confirmed.

27. **e. All of the above.** CO_2 is the most commonly used insufflant for laparoscopic surgery and is favored by most laparoscopists thanks to its properties (colorless, noncombustible, very soluble in blood, and inexpensive).
28. **c. Pulmonary disease.** Helium is an inert and noncombustible insufflant. Initial studies performed in various animal models showed favorable effects on arterial partial pressure of CO_2 and pH with no evidence of hypercarbia. These results were corroborated by clinical studies. Therefore helium is particularly useful for the patient with pulmonary disease in whom hypercarbia would be poorly tolerated.
29. **e. a and b.** If entry into the bowel is not recognized at the time of irrigation and aspiration through the Veress needle, then the surgeon may insufflate the small or large bowel. The first sign of this problem is asymmetric abdominal distention followed by flatus and insufflation of only a small amount of CO_2 (< 2 L) before high pressures are reached.
30. **a. Increase in end-tidal CO_2 .** The diagnosis of CO_2 gas embolism is usually made by the anesthesiologist based on an abrupt increase of end-tidal CO_2 accompanied by a sudden decline in oxygen saturation and then a marked decrease in end-tidal CO_2. Sometimes, a “millwheel” precordial murmur can be auscultated. In addition, the anesthesiologist may notice foaming of a blood sample, if drawn, owing to the presence of insufflated CO_2.
31. **b. Decrease the operating time and improve the operative performance of surgical trainees with limited laparoscopic experience when compared with no training or with box-trainer training.** VR trainers are computer-based simulators that offer the opportunity to practice laparoscopic and robotic skills through specific tasks, as well as whole procedures. VR trainers have been shown to improve the skills of trainees helping to

prepare them for better performance during live surgery. A recent systematic review demonstrated that VR training appears to decrease the operating time and improve the operative performance of surgical trainees with limited laparoscopic experience when compared with no training or with box-trainer training.

32. **c. Age less than 45 years.** Male patients with a BMI greater than 25 undergoing laparoscopic surgery in the lateral position with the kidney rest elevated and the table completely flexed are at highest risk of developing rhabdomyolysis due to flank pressure.
33. **d. Vascular (renal artery or vein).** Various stapling devices are available for tissue occlusion and division. Each staple load cartridge is color-coded depending on the size of the staples: 2.0-mm staples (gray) or 2.5-mm staples (white) are preferred for vascular (renal vein or renal artery) stapling, whereas 3.8-mm (blue) and 4.8-mm (green) staples are used in thicker tissues (ureter, bowel, bladder).
34. **b. Endo Stitch.** The Endo Stitch (Covidien, Mansfield, MA) device is an innovative, disposable 10-mm instrument that facilitates laparoscopic suture placement and knot tying, not port site closure.
35. **a. Incomplete circumferential dissection of the vessel.** The basic principles of Hem-o-Lok (Teleflex, Morrisville, NC) placement include the following:
    - Complete circumferential dissection of the vessel
    - Visualization of the curved tip of the clip around and beyond the vessel, often with curved end of the clip placed between artery and vein
    - Confirmation of the tactile snap when the clip engages
    - No cross clipping
    - Not squeezing clip handles too hard (compared with the application of metal clips)
    - Careful removal of the applier after application given; the tips are sharp and can cause a laceration of nearby vessels (e.g., renal vein)
    - During transection of vessels, only a partial division is performed initially to confirm hemostasis before complete transection

- Minimum of two clips placed on the patient side of the renal hilar vessel

36. **c. Subcutaneous emphysema.** Once the balloon cannula is positioned in the abdominal cavity, the balloon is inflated; the cannula is pulled upward until the balloon is snug on the underside of the abdominal wall. Next, the soft foam or rubber collar on the outside surface of the cannula is slid down until it is snug on the skin and locked in place. This process creates an excellent seal, precluding gas leakage and subcutaneous emphysema.
37. **e. All of the above.** Several actions can be taken by the surgeon to lessen the risks of a thermal complication. First, electrosurgical instruments must be carefully inspected before use for any "breaks" in the insulation; if these are found, the instrument must be sent for recoating. Second, electrosurgical instruments should never be left untended within the abdomen; when not in use, they must be removed from the abdomen. Third, only the primary surgeon should control electrode activation. Fourth, isolation of the area to be cauterized from the surrounding tissues, as well as use of bipolar electrocautery, reduces the risk of thermal spread and injury to other tissues. Fifth, the electrosurgical device should never be activated unless the entire extent of the metal portion of the instrument is in view. Sixth, problems of capacitive coupling can be precluded by not creating a situation in which a mixture of conducting and nonconducting elements are used by the surgeon (e.g., metal trocars combined with plastic retainers). Last, an active electrode monitoring system (Encision, Boulder, CO) is extremely helpful, as any sudden break in the insulation of the electrosurgical instrument results in immediate shutdown of the electrosurgical current, thereby precluding an electrosurgical injury.

Chapter Review

1. Careful patient selection and identification of contraindications for laparoscopic and robotic procedures are vital for successful outcomes. A meticulous past history, focusing on prior surgeries,

and physical examination, detailing the location and extent of all abdominal scars, are the initial steps in patient evaluation for minimally invasive surgery.

2. Contraindications to laparoscopic or robotic surgery include uncorrectable coagulopathy, intestinal obstruction unless treatment is intended, significant abdominal wall infection, massive hemoperitoneum or hemoretroperitoneum, generalized peritonitis, and suspected malignant ascites.
3. A checklist ensuring that all essential equipment is present and operational should be completed just before initiating the pneumoperitoneum. Additional items to check when using the da Vinci Robotic System include ensuring that all plugs for the console, vision cart, and patient-side cart are plugged into different circuits and that all cables connecting these carts are connected properly.
4. After placement of the Veress needle, insufflation should never be initiated unless all of the signs for proper peritoneal entry (negative aspiration, easy irrigation of saline, negative aspiration of saline, positive drop test result, and normal advancement test) have been confirmed.
5. The open technique is recommended specifically when extensive adhesions are anticipated.
6. Noncutting dilating trocars have superseded bladed trocars because they are safer. These trocars enter the abdomen by spreading the abdominal wall musculature, rather than cutting it, and therefore there is less chance of injuring an abdominal wall vessel and the resulting entry site is less prone to subsequent herniation.
7. Carbon dioxide is the most commonly used insufflant because it is noncombustible and rapidly absorbed in the blood.
8. Helium is potentially useful for the patient with pulmonary disease in whom hypercarbia would be poorly tolerated.
9. Intra-abdominal pressures during laparoscopy should not be allowed to exceed 20 mm Hg over extended periods, and a working pressure of 10 to 12 mm Hg is recommended.
10. Early in one's experience with laparoscopic and robotic surgery, it is wise to apply the minimally invasive approach to low-risk surgical candidates of normal body habitus.

11. The first sign of gas embolism is an abrupt increase of end-tidal CO_2 accompanied by a sudden decline in oxygen saturation and then a marked decrease in end-tidal CO_2.
12. Electrosurgically induced thermal bowel injury may occur through one of four mechanisms: inappropriate direct activation; coupling to another instrument; capacitive coupling; and insulation failure.
13. Careful planning of trocar placement is essential to avoid crossing swords, striking handles, rollover, and robotic arm collisions.
14. Laparoscopic VR trainers have been shown to improve the skills of trainees, helping to prepare them for better performance during live surgery.
15. VR robotic trainers have been shown to have face, content, and construct validity.

15

Basic Energy Modalities in Urologic Surgery

Michael W. Sourial, Shubha K. De, Manoj Monga, and Bodo E. Knudsen

Keywords

bipolar; cautery; electrosurgery; Holmium:YAG; intracorporeal lithotripsy; laser; monopolar

Questions

1. What is the mechanism electrosurgery uses to affect tissues?
    a. Current is delivered to the tip of the instrument, causing it to heat and affect the tissue.
    b. Current is delivered to the tissue directly, causing it to heat.
    c. Current is conducted through a fluid medium to affect the tissue.
    d. Electrons are excited, creating increased light energy, which directly affects the tissue.
2. What is the mechanism electrocautery uses to affect tissues?
    a. Current is delivered to the tip of the instrument, causing it to heat and affect the tissue.
    b. Current is delivered to the tissue directly, causing it to heat.
    c. Current is conducted through a fluid medium to affect the tissue.
    d. Electrons are excited, creating increased light energy, which directly affects the tissue.

3. When cautery is set to “pure cut,” the current is:
    a. interrupted but mainly on.
    b. interrupted but mainly off.
    c. continuous.
    d. continuous but oscillates between high and low voltage.
    e. variable in both intermittency and voltage.
4. An argon beam coagulator:
    a. works by igniting a column of argon gas.
    b. uses an argon laser to diffusely coagulate tissues.
    c. should be used in direct contact with the tissue's surface.
    d. should be used in a dry environment.
    e. uses a column of argon gas that passes over an electrode.
5. Bipolar and monopolar cautery differ in that:
    a. monopolar does not require a dispersive electrode.
    b. monopolar can be used at much higher voltages.
    c. bipolar does not require a dispersive electrode.
    d. bipolar can be used at much higher voltages.
    e. there are no differences.
6. The LigaSure and Gyrus PK both show benefits over the Thunderbeat and Ultrashears in that they:
    a. are able to produce less smoke and keep a clear visual field.
    b. seal vessels faster and with higher burst pressures.
    c. function more reliably in wet environments.
    d. are cheaper and reusable.
    e. show no benefits and are inferior products.
7. The wavelength for the holmium:YAG (Ho:YAG)laser is:
    a. 488 nm—blue; 514 nm—green.
    b. 1064 nm.
    c. 1318 nm.
    d. 2140 nm.
    e. 2640 nm.
8. Stone fragmentation via Ho:YAG lasers occurs by:
    a. cavitation bubble collapse and resulting shock waves.
    b. fluid jets created by rapid heating of the surrounding fluid.
    c. pneumatic activity of the laser tip against the stone.
    d. direct energy absorption.

e. ultrasonic thermal ablation of the stone surface.

9. The major downside of pneumatic lithotripsy in the ureter is:
    a. cost.
    b. ureteral injury.
    c. poor visualization.
    d. stone retropulsion.
    e. all of the above.
10. Which instruments can be used through flexible ureteroscopy?
    a. Electrohydraulic lithotripsy (EHL), pneumatic, ultrasonic
    b. Laser, ultrasonic, combination (pneumatic and ultrasonic)
    c. Ultrasonic, laser, EHL
    d. Combination (pneumatic and ultrasound), laser, EHL
    e. All modalities

Answers

1. **b. Current is delivered to the tissue directly, causing it to heat.** Electrosurgery uses radiofrequency current in the range of 400,000 to 600,000 Hz to pass through tissue and create the desired effects. The generators deliver more than 100 W of power to the tissue at voltages ranging from 100 to 5000 V. While the current is delivered to the tissue, the tissue is heated and the effect occurs. This is in contrast to electrocautery, in which the instrument itself is heated and then applied to the tissue.
2. **a. Current is delivered to the tip of the instrument, causing it to heat and affect the tissue.** In contrast to electrosurgery, where radiofrequency current in the range of 400,000 to 600,000 Hz passes through tissue and create the desired effects, with electrocautery the tip of the instrument is heated and then applied to the tissue to create the desired effect.
3. **c. Continuous.** Pure cut uses continuous delivery, whereas coagulation uses interrupted delivery. Generators will also usually provide "blended" modes that modify the degree of interruption to gain the desired effect.

4. **e. Uses a column of argon gas that passes over an electrode.** The argon beam coagulator works by adding a column of argon gas that passes over the electrode; electrosurgical energy ionizes the argon gas and helps to displace the blood in the surgical field. Because argon is a noble gas, the current from the electrode is effectively transmitted to the underlying tissue.
5. **c. Bipolar does not require a dispersive electrode.** Unlike monopolar systems in which a circuit is created by delivering the energy via an electrode and then removed from the patient using a dispersive electrode (grounding pad), bipolar delivery does not require a dispersive electrode. Rather, the active and return electrodes are integrated in the delivery handpiece. The tissue contained between the electrodes is the target tissue.
6. **b. Seal vessels faster, and with higher burst pressures.** A comparison study comparing the vessel sealing times and thermal spread of two bipolar vessel sealing systems (LigaSure and Gyrus PK), as well as an ultrasonic devise (Ethicon Harmonic Scalpel), was performed. This demonstrated that the two bipolar systems had faster vessel sealing times with higher burst pressures compared with the ultrasonic device. However, the ultrasonic device had less thermal spread and smoke production (Lamberton et al., 2008). [a]
7. **d. 2140 nm.** Ho:YAG laser is a 2140-nm pulsed laser that is used for both soft tissue and lithotripsy applications in urology. The 2140-nm wavelength is strongly absorbed in water, traveling only approximately 0.5 mm in the fluid medium, making it ideal for the urologic environment. Both the argon (488 nm—blue; 514 nm—green) and Nd:YAG (1064 nm, 1318 nm) lasers use two different wavelengths.
8. **d. Direct energy absorption.** Previous laser technologies (Ruby, Nd:YAG) used photoacoustic or photomechanical processes, where light energy created shock waves that fragmented stones. In contrast, the Ho:YAG laser uses photothermal lithotripsy, which involves direct light energy absorption (“photo”) by stone surfaces, causing rapid temperature (“thermal”) increases, before significant heat diffusion can occur. A “Moses effect” occurs by the rapid

vaporization of fluid creating a vapor channel between the fiber tip and stone's surface, allowing for more direct energy transfer. Interstitial water may also become vaporized, leading to fragment ejection; however, these forces are not great enough to directly lead to stone fracture.

9. **d. Stone retropulsion.** Pneumatic lithotripsy uses ballistic forces to transfer kinetic energy from a handheld probe to the stone surface. Repetitive strikes (12 Hz LithoClast, 15 to 30 Hz Electrokinetic lithotripter) from the probe tip act as a jackhammer, fragmenting stones at the point of contact. Stone migration is a significant disadvantage when treating ureteric stones because the probe's ballistic effect can propel stones in capacious ureters into the kidney. Retropulsion has been reported in as much as 10% of distal and 40% of proximal stones treated with pneumatic lithotripsy. In a four-way comparison of intracorporeal lithotripters on iatrogenic urothelial trauma, pneumatic probes were found to be the least traumatic (compared with laser, ultrasonic, and EHL). Pneumatic lithotripters are currently one of the most cost effective because of their durability and use of reusable probes.
10. **c. Ultrasonic, laser, EHL.** Because of their mechanisms of action, EHL probes and laser fibers can fragment stones while being flexed; 200-µm laser fibers can be readily flexed 270 degrees in flexible ureteroscopes, whereas thin EHL probes (1.9 Fr) are flexible enough to reach the lower pole while conducting electrical pulses for spark discharge. Thin ultrasonic probes can be moderately deflected using flexible scopes; however, these wirelike probes lack a lumen for suction and suffer from significant dampening and reduced efficiency with flexion. Any amount of torque applied to pneumatic lithotrites significantly reduces the jackhammer-like movements and reduces fragmentation potential. Similarly, combination lithotrites cannot be flexed, nor are they available in sizes compatible with ureteroscopy.

Chapter Review

1. When alternating current is used, the term for resistance is impedance; impedance increases when charring occurs.
2. Lasers with shorter wavelengths have a much greater amount of scatter than those with longer wavelengths.
3. The depth of tissue penetration for Nd:YAG is 10 mm, KTP 1 to 2 mm, Ho:YAG 0.4 mm, and CO_2 no significant penetration.
4. The electrohydraulic probe produces a spark that creates a shock wave; one should place the probe 1 mm away from the stone.
5. Stone composition and surface characteristics affect the efficiency of EHL, with uric acid stones taking the most time to fragment.
6. The pneumatic device is the only modality that does not cut through wire, such as a safety guidewire or stone basket.
7. The ultrasonic probe results in vibration at the end of the probe that is transferred to the stone, causing it to fracture.
8. Ho:YAG laser stone fragmentation is primarily due to a photothermal effect.
9. Laser lithotripsy produces the smallest fragments and is useful for all stone compositions.
10. The argon beam coagulator works by adding a column of argon gas that passes over the electrode, and then electrosurgical energy ionizes the argon gas and helps to displace the blood in the surgical field. Because argon is a noble gas, the current from the electrode is effectively transmitted to the underlying tissue.
11. Bipolar delivery does not require a dispersive electrode (grounding pad). The active and return electrodes are integrated in the delivery handpiece.
12. The bipolar systems for sealing tissue have faster vessel sealing times with higher burst pressures compared with the ultrasonic device. However, the ultrasonic device has less thermal spread and smoke production.
13. Pneumatic probes are the least traumatic to ureteral tissue compared with laser, ultrasonic, and EHL.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

16

Evaluation and Management of Hematuria

Stephen A. Boorjian, Jay D. Raman, and Daniel A. Barocas

Keywords

hematuria; hemorrhagic cystitis; microhematuria; nephropathy; urinalysis

Questions

1. According to AUA guidelines, microhematuria sufficient to trigger a diagnostic evaluation is defined as:
    a. a positive chemical test (urine dipstick) showing small, moderate, or large blood on one properly collected specimen.
    b. a positive chemical test (urine dipstick) showing small, moderate, or large blood on at least two of three properly collected specimens.
    c. a positive chemical test (urine dipstick) showing large blood on one properly collected specimen.
    d. urine microscopy showing three or more red blood cells per high-powered field on one properly collected urine specimen.
    e. urine microscopy showing three or more RBC/HPF on at least two of three properly collected urine specimens.
2. The likelihood of finding a malignancy in a patient with microhematuria is influenced by all of the following EXCEPT:
    a. age.

b. gender.
c. use of anticoagulants.
d. tobacco use.
e. degree of hematuria.

3. According to AUA guidelines, the proper initial assessment of a 50-year-old patient with asymptomatic microhematuria includes:
   a. blood pressure measurement, serum creatinine level, cystoscopy, and computed tomography (CT) urogram.
   b. urine cytology, cystoscopy, and CT urogram.
   c. urine cytology, blue-light cystoscopy, and any upper tract imaging.
   d. urine cytology and renal/bladder ultrasound.
   e. no evaluation is necessary unless microhematuria is persistent/recurrent or hematuria is visible.
4. In the evaluation of patients with microhematuria, cystoscopy may be safely avoided if:
   a. there are no associated symptoms in a patient of any age.
   b. the patient is under 35 years of age and without symptoms or risk factors for malignancy.
   c. the patient is taking aspirin or warfarin.
   d. the cytology is negative.
   e. the patient has a history of urinary tract infection and hematuria is still present after treatment.
5. Patients presenting with gross hematuria in the absence of recent trauma or concurrent infection who are on anticoagulation medications should be evaluated with:
   a. urinalysis, urine cytology, and cystoscopy only.
   b. CT urogram, with cystoscopy only if symptomatic.
   c. no evaluation necessary.
   d. assessment of anticoagulation status, and evaluation only if supra-therapeutic.
   e. urine cytology, cystoscopy, CT urogram.
6. The metabolite of oxazaphosphorine chemotherapeutic agents which is responsible for hemorrhagic cystitis is:
   a. Mesna.
   b. Acrolein.

c. Formalin.
d. Gemcitabine.
e. Methotrexate.

7. Use of intravesical alum for hemorrhagic cystitis should be avoided in patients with:
   a. a history of malignancy.
   b. a history of detrusor instability.
   c. active gross hematuria.
   d. renal insufficiency.
   e. vesicoureteral reflux.
8. The molecular pathophysiology linking BPH and hematuria is exemplified by the identification of which of the following in the prostate tissue from men with BPH:
   a. decreased microvessel density
   b. androgen-independent angiogenesis
   c. elevated VEGF expression
   d. reduced cell proliferation
   e. diminished blood flow
9. A 65-year-old man with a history of BPH has recurrent gross hematuria. The patient is clinically stable, with no transfusion requirement, no clots in urine, and no difficulty with bladder emptying. A hematuria evaluation with CT urogram, cystoscopy, and urine cytology is otherwise unremarkable. The best next step in management is:
   a. five-alpha reductase inhibitor.
   b. alpha-blocker therapy.
   c. prostatic artery embolization (PAE).
   d. channel TURP.
   e. trial of antibiotic therapy.
10. A 35-year-old man presents with complaint of penile pain and immediate detumescence during intercourse. Physical examination notes blood at the urethral meatus. The next step should be:
   a. immediate operative exploration.
   b. CT scan of pelvis.
   c. retrograde urethrography.
   d. obtain serum coagulation parameters.

e. conservative management with serial examinations.

11. A 55-year-old woman presents with intermittent gross hematuria 2 weeks after undergoing a right partial nephrectomy for a 4-cm solid enhancing renal mass. She is afebrile with stable vital signs. She is able to void to completion and her urine is red without clots. Her creatinine is 1.1 mg/dL. The next step should be:
    a. surgical exploration.
    b. renal angiography.
    c. continuous bladder irrigation.
    d. observation.
    e. noncontrast CT scan of the abdomen/pelvis.

Answers

1. **d. Urine microscopy showing three or more red blood cells per high-powered field on one properly collected urine specimen. The presence of three of more RBCs/HPF on a single urine microscopy is associated with malignancy in 2.3-5.5% of patients.** Chemical tests for hematuria detect the peroxidase activity of erythrocytes using benzidine, and can render false results in the presence of dehydration, myoglobinuria, high doses of vitamin C, improper technique, and other factors. While higher levels of microhematuria (>25 RBCs/HPF) are known to be associated with higher rates of malignancy on evaluation, setting the threshold higher than three RBCs/HPF or requiring more than one positive urinalysis would lead to an unknown number of missed opportunities for diagnosis.
2. **c. Use of anticoagulants. Increasing age, male gender, and tobacco use are risk factors for urologic cancers and specifically for urothelial carcinoma.** In addition, while there is little data to distinguish among thresholds of two, three, four, or five RBCs/HPF, it is clear that a high level of microhematuria (>25 RBCs/HPF) is associated with a greater likelihood of malignancy. By contrast, patients using anticoagulant medications or anti-platelet medications have a similar risk of malignancy compared to

those who do not use these medications. Therefore such patients should be evaluated comparably to those who do not use anticoagulants or anti-platelet agents.

3. **a. Blood pressure measurement, serum creatinine level, cystoscopy, and computed tomography (CT) urogram. The AUA suggests that adult patients presenting with asymptomatic microhematuria should undergo evaluation to determine the cause.** Blood pressure measurement and serum creatinine level may help identify patients who require concurrent nephrologic workup, and creatinine level also helps determine patient eligibility for contrast imaging. The evaluation of asymptomatic hematuria includes imaging (preferably with CT urogram), and cystoscopy in patients 35 and older and those under 35 with risk factors for malignancy.
4. **b. The patient is under 35 years of age and without symptoms or risk factors for malignancy. The AUA guidelines (** Fig. 16.1 **) call for use of cystoscopy for evaluation of hematuria in all patients 35 years of age and older (Recommendation).** The risk of malignancy is very low in persons under 35 years of age, such that the potential benefits of cystoscopy may be outweighed by the very small risks associated with the procedure. Therefore it is an option to omit cystoscopy in patients under the age of 35, provided that the patient does not have risk factors for a urologic malignancy.
5. **e. Urine cytology, cystoscopy, CT urogram. Given the increased frequency with which clinically significant findings are associated with gross hematuria, the recommended evaluation in this setting is relatively uniform.** That is, patients presenting with gross hematuria in the absence of antecedent trauma or culture-documented urinary tract infection should be evaluated with a urine cytology, cystoscopy, and upper tract imaging, preferably CT urogram. Importantly, patients who develop hematuria who are on anticoagulation medications should undergo a complete evaluation in the **same manner as patients not taking such medications, as the prevalence** of hematuria, as well as the likelihood of finding genitourinary cancers, among patients with

hematuria on anticoagulation has been reported to be no different from patients not taking such medications.

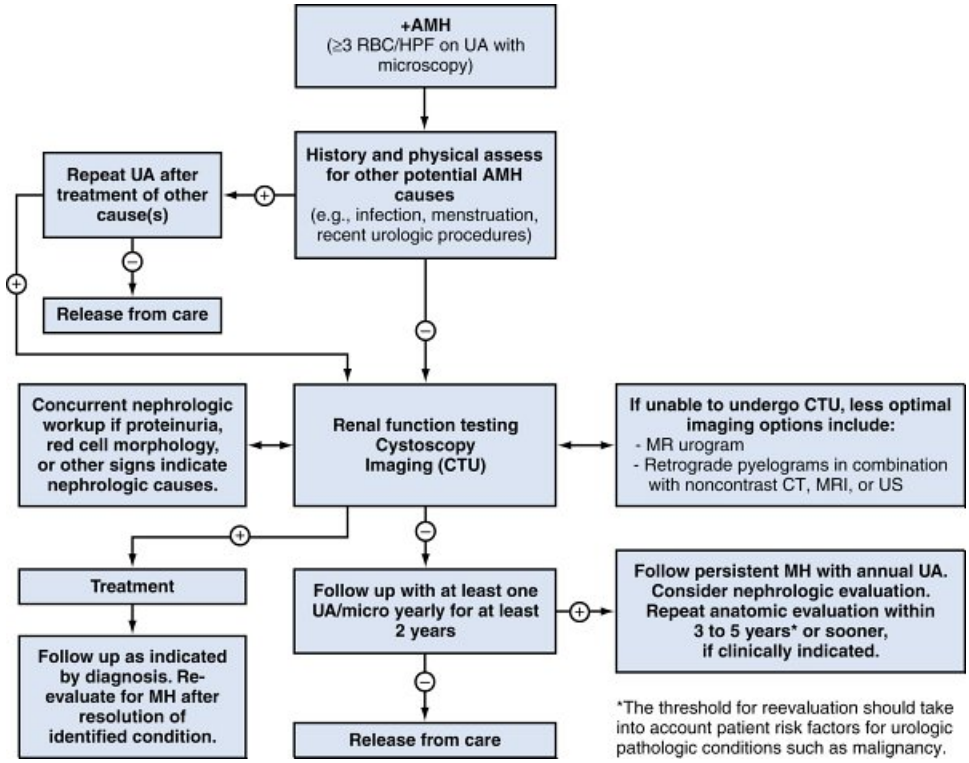


FIG. 16.1 American Urological Association guideline algorithm for evaluation of adult patients with asymptomatic microhematuria. *AMH*, Asymptomatic microhematuria; *CT*, computed tomography; *CTU*, computed tomography urogram; *HPF*, high-power field; *MH*, microhematuria; *MR*, magnetic resonance; *MRI*, magnetic resonance imaging; *RBC*, red blood cell; *UA*, urinalysis; *US*, ultrasound.

6. **b. Acrolein. Bladder toxicity from oxazaphosphorine chemotherapeutic agents results from renal excretion of the metabolite acrolein, which is produced by the liver and which stimulates bladder mucosal sloughing and subsequent tissue edema/fibrosis.** Mesna (2-mercaptoethane sulfonate), which binds to acrolein and renders it inert, has been suggested for prophylaxis against cyclophosphamide-induced hemorrhagic cystitis.
7. **d. Renal insufficiency. Alum may be considered for first-line intravesical therapy among patients with hemorrhagic cystitis (**

Fig. 16.2) **failing initial supportive measures.** However, while cell penetration and therefore overall toxicity of this agent are low, systemic absorption may nevertheless occur and may result in aluminum toxicity, with consequent mental status changes, particularly among patients with renal insufficiency. Meanwhile, prior to intravesical administration of silver nitrate or of formalin, a cystogram should be obtained to evaluate for the presence of vesicoureteral reflux.

8. **c. Elevated VEGF expression. The etiology for BPH-related hematuria has been thought to be increased prostatic vascularity due to higher microvessel density in hyperplastic prostate tissue.** This noted increase in microvessel density has in turn been linked to higher levels of VEGF. Moreover, as the pathophysiology of BPH-related bleeding has been postulated as increased cell proliferation stimulating increased vascularity, efforts to suppress prostate growth via androgen ablation have been explored. Indeed, both estrogens and antiandrogens have, in small case reports, been associated with decreased prostate bleeding, presumably through the repression of androgen-stimulated angiogenesis and the induction of programmed cell death within the prostate.
9. **a. Five-alpha reductase inhibitor. Treatment with five-alpha reductase inhibitors has been associated with decreased VEGF expression, prostate microvessel density, and prostatic blood flow.** Clinically, multiple series have demonstrated efficacy of finasteride for BPH-related hematuria, with symptom improvement or resolution consistently noted in approximately 90% of patients. As such, in otherwise stable patients, finasteride represents a reasonable first-line therapy for BPH-related gross hematuria after the completion of an initial diagnostic evaluation. Channel TURP has typically been utilized in the setting of prostate cancer, while a "standard" TURP or an alternative form of such endoscopic prostate tissue removal/destruction may be used for patients with persistent bleeding from BPH despite conservative therapies and/or endoscopic fulguration, particularly when additional indications for BPH surgery coexist. In cases with persistent bleeding despite

TURP or in the significantly comorbid patient, prostatic artery embolization can be considered.

10. **c. Retrograde urethrography. The clinical scenario is consistent with an acute penile fracture.** Blood at the meatus raises suspicion for concomitant urethral injury that requires investigation by retrograde urethrography prior to planned operative repair. Conservative management poses the risk of untreated urethral and corporal injury that can result in urethral stricture disease and erectile dysfunction.
11. **b. Renal angiography. The clinical scenario is consistent with a renal arteriovenous malformation (AVM).** Renal angiography may be both diagnostic and therapeutic in this scenario with ability to coil or embolize this abnormal vascular communication. Observation and bladder irrigation do not address the underlying causative factor while non-contrast CT imaging fails to delineate the vascular anatomy. Surgical exploration has a high likelihood of renal loss and is reserved for cases refractory to angiographic modalities.

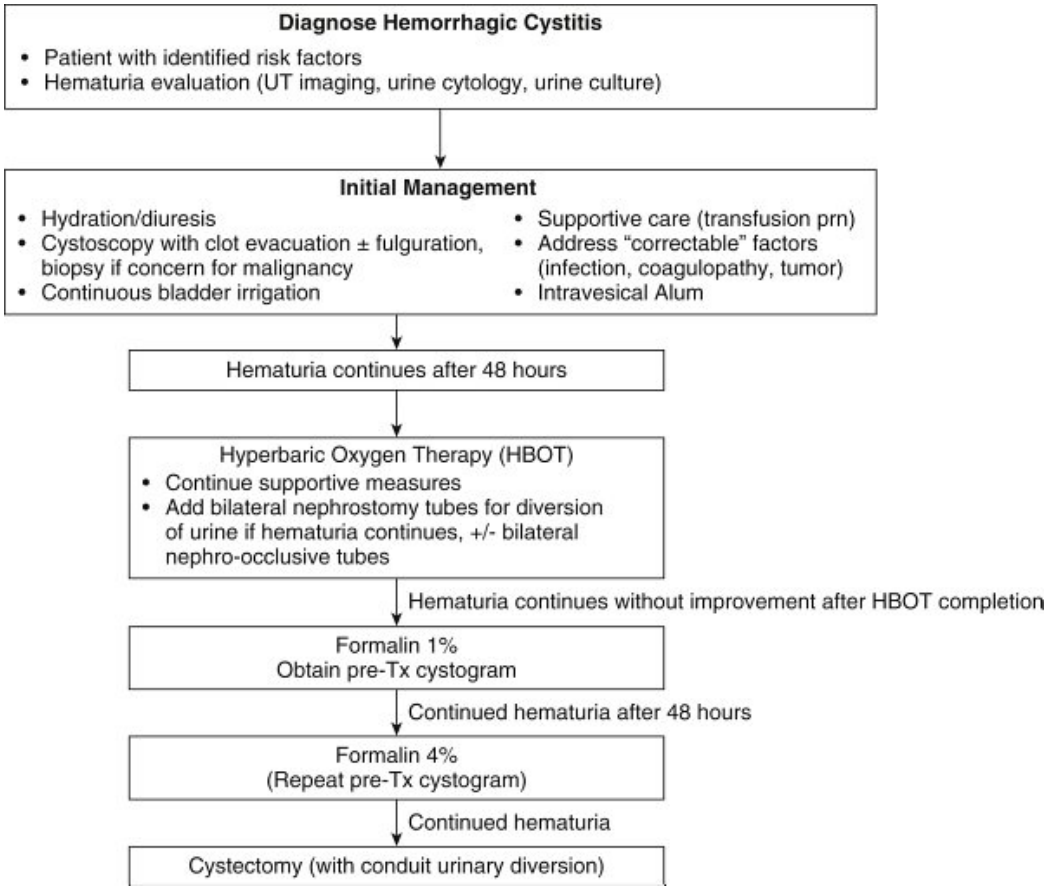


FIG. 16.2 Management algorithm for patients with hemorrhagic cystitis. *HBOT*, Hyperbaric oxygen therapy; *Tx*, treatment; *UT*, upper tract.

Chapter Review

1. Microscopic hematuria (MH) is defined by the AUA guidelines as three or more RBCs/HPF, identified on one or more occasions on urine microscopy. Urine dipstick testing is insufficient for the diagnosis of MH.
2. MH is quite common, with a prevalence of approximately 6.5% of adults, varying according to the characteristics of the population.
3. Malignancy has been detected in approximately 3.5% of patients evaluated for asymptomatic MH. The risk of malignancy diagnosis is greater in patients with higher degrees of hematuria, male gender,

and/or risk factors for malignancy, and lower in absence of these risk factors.

4. Evaluation of adults with microscopic hematuria includes a history and physical examination, renal function testing, and upper tract imaging for all patients.
5. White light cystoscopy is recommended in the evaluation of asymptomatic MH for patients 35 years of age or older and/or those with risk factors for malignancy.
6. CT urogram is the preferred imaging modality for the evaluation of hematuria.
7. Urine cytologic examination and biomarkers are not indicated in the initial evaluation of asymptomatic MH.
8. Patients with a negative complete evaluation can be released from care if subsequent urinalyses confirm resolution of MH. Reevaluation should be considered in patients with persistent/recurrent MH and those with an incomplete initial evaluation.
9. Oxazaphosphorine chemotherapeutic agents have been linked to the development of hemorrhagic cystitis through exposure of the metabolite acrolein to the urothelium.
10. Alum may be used as a first-line intravesical therapy for hemorrhagic cystitis in patients without renal dysfunction.
11. Formalin is a highly effective form of intravesical therapy for hemorrhagic cystitis. A cystogram should be obtained before therapy to ensure no vesicoureteral reflux.
12. HBOT has been associated with response rates of 80% to 100% for patients with hemorrhagic cystitis.
13. BPH represents the most common cause of GH in men older than 60 years.
14. 5α-Reductase inhibitors may be used for BPH-related GH, with potential improvement having been noted within weeks after instituting therapy.
15. Androgen deprivation may be effective for patients with locally advanced prostate cancer with GH.
16. Angioembolization and/or urinary diversion represent salvage options for management for patients with refractory hematuria,

pending clinical status.

17. Urethral bleeding should be suspected with blood at the meatus and/or initial hematuria.
18. A concern for traumatic urethral injury should prompt retrograde urethrogram.
19. Urinary findings suggestive of a glomerular or tubulointerstitial medical renal disease include the presence of RBC casts in the urinary sediment, dysmorphic RBCs, and proteinuria.
20. In patients with GH after a recent renal procedure, expeditious angiography should be considered to allow for the diagnosis and management of renal AVM.

17

Complications of Urologic Surgery

Reza Ghavamian, and Charbel Chalouhy

Keywords

complications; injury; laparoscopy; robotic surgery; urologic surgery; urology

Questions

1. During posterior dissection in a robotic salvage radical prostatectomy, an inadvertent rectal injury is encountered. Gross fecal spillage is seen in the surgical field. What does a good management strategy include?
    a. Closure of rectal mucosa and serosa with non-absorbable sutures
    b. Open conversion, colostomy, and delayed repair
    c. Broad-spectrum antibiotics are advised for up to 1 month postoperatively
    d. Primary closure and omental flap
    e. Open approach to radical prostatectomy is preferable in this case given the lower incidence of injuries
2. Which of the following has the greatest effect on pseudoaneurysm development following partial nephrectomy?
    a. The open approach versus minimally invasive (1% vs. 5%)
    b. Patients with higher nephrometry score are at increased risk
    c. Early unclamping even in the presence of arterial bleed

d. Tumor depth has no impact on incidence of pseudoaneurysm
e. Deep suture repair through the renal parenchyma is the best preventative measure

3. When obtaining pneumoperitoneum with the VERESS needle:
    a. the angle of entry for the needle should be near 90 degrees for people with normal body mass index (BMI).
    b. the angle of entry for the needle should be near 45 degrees for people with high BMI.
    c. the most commonly injured vessel is the right iliac vein.
    d. the most commonly injured vessel is the left common iliac vein.
    e. this access modality has been increasingly abandoned and replaced by optical trocars.
4. What is true about urine leaks after partial nephrectomy?
    a. They are more likely to cause ileus during open surgery
    b. 90% of patients will require readmission.
    c. The mean postoperative day for presentation is 18 days.
    d. Longer warm ischemia times are directly related to the development of urine leaks.
    e. This complication is more common in minimally invasive surgery.
5. Once all attempts for conservative management have failed for ureteral complications, it is agreed that:
    a. waiting 6 weeks for surgical repair allowing inflammation to subside is reasonable.
    b. early surgical repair can be achieved when discovered early within 3 weeks.
    c. urinomas do not have to be drained to prevent re-accumulation.
    d. there is no increased risk for stricture.
    e. robotic approach for Boari flap and psoas hitch should be avoided.
6. During Psoas hitch procedure, the most commonly injured structure is:
    a. obturator nerve.
    b. ilio-inguinal nerve.

c. pudendal nerve.
d. sympathetic nerves.
e. genito-femoral nerve.

7. Brachial plexus injuries can be prevented by:
    a. adequate padding of the arms.
    b. arm positioning in external rotation and posterior shoulder displacement.
    c. avoiding excess arm abduction of greater than 90 degrees.
    d. preferring modified lateral to lateral positions.
    e. limiting axillary rolls.

Answers

1. **b. Open conversion, colostomy, and delayed repair.** A good management strategy in this case includes open conversion, colostomy, and delayed repair.
2. **b. Patients with higher nephrometry score are at increased risk.** Patients with higher nephrometry score are at increased risk for pseudoaneurysm development following partial nephrectomy.
3. **d. The most commonly injured vessel is the left common iliac vein.** When obtaining pneumoperitoneum with the Veress needle, the most commonly injured vessel is the left common iliac vein.
4. **d. Longer warm ischemia times are directly related to the development of urine leaks.** Longer warm ischemia times are directly related to the development of urine leaks after partial nephrectomy.
5. **a. Waiting 6 weeks for surgical repair allowing inflammation to subside is reasonable.** Once all attempts for conservative management have failed for ureteral complications, it is agreed that waiting 6 weeks for surgical repair allowing inflammation to subside is reasonable.
6. **e. Genito-femoral nerve.** The most commonly injured structure during Psoas hitch procedure is the genito-femoral nerve.
7. **c. Avoiding excess arm abduction of greater than 90 degrees.** Avoiding excess arm abduction of greater than 90 degrees can

prevent brachial plexus injuries.

Chapter Review

1. A wide array of physiologic, urinary, vascular, bowel, and visceral complications can occur with all forms of urologic operations.
2. With the wide adoption of laparoscopic and robotic surgery in urology, unique sets of complications are encountered.
3. As more complex operations are performed in a minimally invasive fashion, the rate of complications can increase.
4. Some complications after laparoscopic and robotic procedures can present in an atypical and delayed fashion.
5. A high index of suspicion is necessary for prompt and timely intraoperative and postoperative diagnosis and management.
6. The most important aspect of complication management is complication prevention with adequate planning, attention to detail, and meticulous surgical technique.

18

Urologic Considerations in Pregnancy

Melissa R. Kaufman

Keywords

bladder exstrophy; hydronephrosis; myelomeningocele; placenta percreta; pregnancy; urinary physiology; urinary tract infection; urolithiasis

Questions

1. Decreased cardiac output in pregnant patients undergoing urologic surgery may be ameliorated by the following maneuver:
    a. Aggressive intraoperative fluid resuscitation
    b. Administration of alpha receptor agonists
    c. Use of only regional anesthesia
    d. Right-side elevation allowing displacement of the uterus off the vena cava
    e. High lithotomy position to increase venous return from lower extremities
2. Changes in renal tubular function during pregnancy result in alterations of urine chemistries EXCEPT for:
    a. glycosuria.
    b. proteinuria.
    c. pyuria.
    d. albuminuria.

e. calcinuria.

3. Symptomatic physiologic hydronephrosis of pregnancy should be primarily managed by:
    a. percutaneous nephrostomy.
    b. conservative measures.
    c. antibiotic prophylaxis.
    d. ureteral stent.
    e. Alpha-blocker therapy.
4. Which of the following statements is FALSE regarding imaging in pregnant patients?
    a. Ultrasound and magnetic resonance imaging are the techniques of choice.
    b. Routine imaging with magnetic resonance imaging (MRI) should employ use of gadolinium contrast.
    c. Radiation from computed tomography (CT) scan is usually at a dose lower than that associated with fetal harm.
    d. Safety of nuclear medicine studies is dependent on the isotope utilized.
    e. None of the above.
5. Current practice recommendations for pregnant women undergoing nonobstetric surgery include:
    a. Anesthetic agents should be avoided due to teratogenic effects.
    b. Elective surgery is best accomplished in the first trimester.
    c. Use of fetal heart rate monitoring during the first trimester to influence delivery decisions.
    d. Surgery should be performed at an institution with neonatal and pediatric services.
    e. None of the above.
6. Which of the following statements regarding stone disease in pregnancy is TRUE?
    a. Approximately 30% of pregnant women with symptomatic calculi pass the stone spontaneously.
    b. Stone events in pregnancy carry an increased risk of maternal and fetal morbidity.
    c. Complication rates of ureteroscopy in pregnant women is 27%.

d. Ureteroscopy and stent placement should always be performed without fluoroscopic guidance.
e. None of the above.

7. Counseling of female patients with myelomeningocele (MMC) desiring pregnancy includes:
   a. Women with MMC in general are considered to have normal fertility.
   b. There is no risk of genetic transmission of neural tube defects to the offspring.
   c. Patients with preexisting renal insufficiency do not have increased risk during pregnancy of worsening renal function.
   d. Women with MMC do not require folic acid supplementation during conception.
   e. None of the above.
8. All of the following regarding placenta percreta are correct EXCEPT:
   a. Prior cesarean section is the predominant risk factor for abnormal placentation.
   b. Biopsy prior to surgical intervention is indicated for placenta percreta.
   c. A multidisciplinary team is mandated to deliver specialized care.
   d. Preoperative counseling should include discussion of ureteral catheters, stents, urinary tract injury, or possible extirpative surgery with urinary diversion.
   e. None of the above.

Answers

1. **d. Right-side elevation allowing displacement of the uterus off the vena cava.** CO in pregnancy is dependent on maternal position and standard techniques for urology may pose unintended consequences. In the supine position, CO in the pregnant woman decreases by up to 30% at term as compared to a lateral recumbent position. **Prevention of potential acute drops in venous return**

and CO may require avoidance of supine and lithotomy positions when feasible with right-side elevation to displace the uterus to the left. Placement of a wedge may be logistically challenging but achieving a 15- to 20-degree left tilt should reduce uterine compression of the inferior vena cava.

2. **c. Pyuria.** Marked change in renal tubular function during pregnancy results in a variety of alterations of urine chemistries that may be encountered by the urologist although elevation of ketones is not typical. Indeed, glycosuria during pregnancy is exceedingly common, with 10- to 100-fold increase in urinary glucose excretion described. Proteinuria and albuminuria additionally manifest during pregnancy and may increase as the gestation advances. Calcinuria doubles during pregnancy. Although women may display pyuria due to contamination or superimposed infection, it is not a consistent finding during gestation.
3. **b. Conservative measures. In general, ureteral stent insertion in symptomatic gestational hydronephrosis does not generally add benefit over conservative treatment and should be reserved for complicated cases or additional pathologic obstructions.** Conservative measures include analgesia, position, and intravenous fluids and failure included signs of evolving infection or deteriorating renal function. Parameters that may increase the likelihood of surgical treatment with ureteral stenting for symptomatic hydronephrosis include maximal anterior-posterior renal pelvis diameter, parity, C-reactive protein level, white blood cell count, and pain on a visual analog scale. Rarely is percutaneous nephrostomy required, and there is not a role for antibiotics outside of acute infection. Alpha blockers are likewise not indicated for relief of colic from physiologic hydronephrosis.
4. **b. Routine imaging with magnetic resonance imaging (MRI) should employ use of gadolinium contrast.** This statement is false. The American College of Obstetricians and Gynecologists (ACOG) have outlined clinical guidelines to drive management decisions and determine the diagnostic imaging modality which provides the optimal risk/benefit ratio in many situations. The use of gadolinium contrast with MRI should be limited; it may be used

as a contrast agent in the pregnant woman only if it significantly improves diagnostic performance and is expected to improve fetal or maternal outcomes.

5. **d. Surgery should be performed at an institution with neonatal and pediatric services.** Recommendations from the ACOG provide the urologist guidance regarding considerations for nonobstetric surgery during pregnancy. Overall, consultation with an obstetrician should be obtained to ensure pertinent aspects of maternal physiology are considered along with potential need for fetal monitoring. **If possible, nonurgent surgery should be performed in the second trimester when preterm contractions and spontaneous abortion are least likely.**
6. **b. Stone events in pregnancy carry an increased risk of maternal and fetal morbidity.** Of pregnant patients with symptomatic calculi, 50% to 80% will pass their stones spontaneously when treated conservatively with hydration and analgesia. However, a stone event in pregnancy does carry with it an increased risk of maternal and fetal morbidity, so patients should be followed closely for recurrent or persistent symptoms. Ureteroscopy in pregnancy carries an overall complication rate of 8.3%, with no significant difference in rates of urinary tract complications compared to the non-pregnant population. Although methods to reduce fetal radiation exposure should be employed, rarely would endoscopic procedures need to be performed without the aid of fluoroscopy.
7. **a. Women with MMC in general are considered to have normal fertility.** Females with MMC are considered to have normal fertility with up to 70% of those who conceive having successful pregnancies. With regard to transmission, the threat is identical if the affected parent is male or female; however, the incidence in female offspring is 1 in 13 and diminishes to 1 in 50 for male children. However, if both parents are affected, the neural tube defect transmission rate increases dramatically to 15%. Foremost is recognition that **the greatest risk factor for worsening renal function in pregnant women with or without urinary reconstruction is preexisting renal insufficiency.** Notably,

compared to routine supplementation of women without MMC of 0.4 mg folic acid per day, **doses of 4 to 5 mg/day have been advocated for the MMC patient desirous of fertility.**

8. **b. Biopsy prior to surgical intervention is indicated for placenta percreta.** With knowledge of the potential risk factors for abnormal placentation, the clinician may choose to proceed with cystoscopic evaluation of such patients to evaluate the posterior bladder wall and significant invasion. Biopsy of the hypervascular tissue visualized in the bladder is discouraged. Assembly of a multidisciplinary team is currently recommended standard of care to reduce maternal and fetal morbidity and mortality. Suggested involvement includes an appropriate anesthesiologist, obstetrician, experienced pelvic surgeon such as a gynecologic oncologist, intensivist, maternal-fetal medicine specialist, interventional radiologist, and as required additional surgical subspecialties such as urology and general surgery. **Preoperative planning by the urologist should include appropriate counseling of the patient with regard to placement of ureteral catheters or stents as well as potential manifestations of substantial injury or need for extirpative surgery for urologic organs and urinary diversion.** Prior cesarean delivery appears to be the predominant risk factor due to abnormal implantation in the myometrial scar with compromised vascularity. Advanced maternal age is additionally contributory with women over 35 years of age accounting for 57% of placental adhesive disorders.

Chapter Review

1. Management of symptomatic physiologic hydronephrosis in the absence of other pathologies such as calculi remains a dilemma for the consulting urologist. Ureteral stent insertion in symptomatic gestational hydronephrosis does not generally add benefit over conservative treatment and should be reserved for complicated cases or additional pathologic obstructions.

2. Ultrasonography and MRI are the imaging techniques of choice for the pregnant patient, but they should be used prudently and only when use is expected to answer a relevant clinical question or otherwise provide medical benefit to the patient. With few exceptions, radiation exposure through radiography, CT scan, or nuclear medicine imaging techniques is at a dose much lower than the exposure associated with fetal harm. If these techniques are necessary in addition to ultrasonography or MRI or are more readily available for the diagnosis in question, they should not be withheld from a pregnant patient.
3. A pregnant woman should never be denied indicated surgery, regardless of trimester. If possible, nonurgent surgery should be performed in the second trimester when preterm contractions and spontaneous abortion are least likely.
4. Due to the tendency for the gravid uterus to compress the IVC, optimizing venous return often requires placing the pregnant surgical patient in a left lateral decubitus position when feasible. This left lateral position should be modified and considered during open, laparoscopic, and endoscopic procedures.
5. Of pregnant patients with symptomatic calculi, the majority will pass their stones spontaneously when treated conservatively with hydration and analgesia. However, for patients requiring intervention, ureteroscopy may be performed with no significant difference in rates of urinary tract complications compared to the non-pregnant population.
6. Most patients with congenital conditions affecting the lower urinary tract have normal fertility and should have prenatal counseling. Complexities of pregnancy in the population with prior urinary diversion require an individualized approach. Mode of delivery is dependent on prior urinary reconstructions and patient preference, although in most cases cesarean section is preferred for the bladder exstrophy patient.
7. Since pregnant women with asymptomatic bacteriuria (ASB) remain at higher risk for development of symptomatic lower or upper urinary tract infections that may precipitate adverse fetal outcomes such as decreased birth weight and premature labor, consensus

opinions indicate that pregnant women should be screened for ASB and positive cultures should be treated.

8. Women diagnosed antenatally with placenta percreta should be referred to a tertiary care center and managed by a multidisciplinary team that includes urologists, general surgeons, obstetric anesthesiologists, interventional radiologists, and intensivists. Urologists should carefully counsel patients regarding complexities of management and outcomes for urinary tract injury including possible urinary diversion.

19

Intraoperative Consultation

Michael J. Schwartz, and Jessica E. Kreshover

Keywords

bladder injury; intraoperative consultation; intraoperative decision making; ureteral injury; urethral injury; urology consult

Questions

1. A colorectal surgeon calls you for intraoperative consultation. She believes that the right ureter "looks denuded" and wants your opinion on management. There is no prior history of radiation therapy to the abdomen or pelvis. You scrub into the surgery, inspect the field, and find a partial-thickness injury to approximately 1⁄3 of the circumference of the distal right ureter. No free fluid is present. The surgeon states that cautery may have been used in the area during dissection but is not sure. Retrograde pyelogram shows no extravasation. The best option for management is:
    a. perform cystoscopy and insertion of right ureteral stent.
    b. perform primary suture imbrication of the injured area and cover with omentum.
    c. debride and reimplant the right ureter with or without a Boari flap or psoas hitch.
    d. remove the segment of injured ureter and perform ureteroureterostomy.

e. replacement of the right distal ureter with ileum.

2. A bladder injury has been identified by an obstetrician immediately after tertiary cesarean delivery and you are called to evaluate and repair the injury intraoperatively. You arrive 20 minutes later, and when you arrive, the obstetrician shows you the 2-cm cystotomy that he closed while waiting for you. He reports that a catheter tip (14 F) had been visible through a small cystotomy at the anterior bladder dome. Your next step is:
    a. dictate an intraoperative consultation stating that the bladder had been injured and repaired by the obstetrician and no further urologic evaluation was required.
    b. leave a pelvic drain and larger catheter in place.
    c. cystoscopy to inspect the closure.
    d. scrub in to explore the bladder and visually inspect the closure.
    e. reopen the cystotomy and enlarge it as needed to inspect the ureteral orifices before closing and irrigating the bladder.
3. Your former co-resident calls by phone after causing an avulsion injury of the proximal left ureter during ureteroscopy for a ureteropelvic junction (UPJ) stone. The patient is stable. Your colleague is proposing a laparoscopic nephrectomy while the patient is under anesthesia, as he is concerned that repairing the ureter would require open surgery and possibly fail. He is very uncomfortable with performance of a primary repair himself and concerned about possible stricture development. He is on call and his partner, whom he would typically ask to help, is out of town until tomorrow. He asks if you agree with nephrectomy. You:
    a. disagree. He should perform primary ureteroureterostomy over a stent.
    b. disagree. He should discuss options with the family including primary repair, ureteral replacement (ileal ureter), autotransplant, and nephrectomy. The family should make the decision and he should proceed with their choice.
    c. disagree. He should speak with the family, temporize the patient with a nephrostomy, awaken the patient, and wait for

available colleague comfortable with ureteroureterostomy to assist.
   d. agree. Nephrectomy holds no risk of stricture or other reparative procedures. It is the best and lowest-risk option.
   e. agree, but counsel him to present this to the patient's family as the best option before proceeding.

4. Gross hematuria is noted in the catheter drainage bag by the anesthesiologist just after a colorectal surgeon has completed a proctocolectomy for Crohn's disease and creation of a J-pouch. The 43-year-old male patient's abdomen is closed. You arrive to find the surgical drapes taken down, the patient in lithotomy still under anesthesia, and a catheter in place draining red urine. After prepping and draping the patient, you perform a cystoscopy and find two bleeding areas that appear to be at sites of perforations along the posterior bladder wall. After cauterizing the bleeding to improve your vision, you inspect the bladder and find no other injury. Your next step is to:
   a. perform bilateral retrograde studies and a cystogram.
   b. inject biologic glue via the cystoscope at the two sites and leave a large catheter for drainage.
   c. reopen the abdomen and repair the bladder perforations.
   d. leave large catheter and start continuous bladder irrigation.
   e. reopen the abdomen, repair the bladder, and dissect out the ureters bilaterally for visual inspection.

5. During a laparoscopic hysterectomy for a large benign fibroid uterus, the distal right ureter is cauterized with a bipolar cautery instrument and appears segmentally devascularized. There is no free fluid in the field. The patient is stable. After performing cystoscopy and left retrograde pyelogram to assess for bladder and contralateral ureteral injury, you find no other abnormality and recommend right ureteral reimplant. However, the gynecologist is strongly opposed to reimplant and insists on stent insertion, as the continuity of the ureter has not been disrupted. What is your next step?
   a. Defer to the primary surgeon, as she has a preexisting relationship with the patient and retains ultimate decision-

making authority, even if you disagree.

b. Offer the option of ureteroureterostomy over a stent, which would eliminate the need for the reimplant the gynecologist wants to avoid.
c. Do nothing immediately, but defer and awaken the patient to allow her to participate in the decision for stent or reimplant.
d. Perform the ureteral reimplant, involving the hospital medical director if necessary. The reimplant is in the patient's best interest, and deferring the decision is likely in this case only to cause harm.
e. Agree to the stent, but insist that it remain in place for 6 to 8 weeks, as you are concerned about stricture or fistula formation otherwise.

Answers

1. **c. Debride and reimplant the right ureter with or without a Boari flap or psoas hitch.** Although there is no extravasation, with a potential thermal injury, the extent of the injury may not be immediately apparent. As injury is within the distal 1⁄3 of the ureter, reimplantation should be undertaken to minimize the much greater potential morbidity associated with delayed recognition.
2. **e. Reopen the cystotomy and enlarge it as needed to inspect the ureteral orifices before closing and irrigating the bladder.** Need to ensure appropriate closure as well as inspect for concomitant injuries. Relying upon the other surgeon to complete the assessment is not meeting the expected role of the consultant.
3. **c. Disagree. He should speak with the family, temporize the patient with a nephrostomy, awaken the patient, and wait for available colleague comfortable with ureteroureterostomy to assist.** It is important for surgeons to balance recognition of their own limitations with prioritizing the best interests of the patient. In this case, there is an available temporizing measure that will allow the surgeon to act in the best interest of the patient and avoid unnecessary risk to the patient by performing a surgery outside of

his own skillset. Performing a nephrectomy would not be the standard of care in a patient with presumably normal renal function being treated for a ureteral stone.

4. **a. Perform bilateral retrograde studies and a cystogram.** As there is potential need for reopening the patient and repairing the bladder in this case, performance of imaging to assess the ureters and bladder is a critical step that will direct the extent of exploration required. If the retrograde studies are normal, this will remove the need for ureteral exploration and minimize the time for any additional repair needed. The cystogram would also help determine if the bladder injury is extraperitoneal or intraperitoneal, influencing the decision for reopening and repairing the bladder.
5. **d. Perform the ureteral reimplant, involving the hospital medical director if necessary. The reimplant is in the patient's best interest, and deferring the decision is likely in this case only to cause harm.** The decision maker for an intraoperative consultation is the consultant. In this instance, knowing the potential harm that may occur with a delayed injury, it is in the patient's best interest to perform ureteral reimplant.

Chapter Review

1. The intraoperative consultation enters the physician into a new patient-physician relationship, and full disclosure regarding the reason for consultation is critical.
2. When possible, all relevant stakeholders should be included when difficult intraoperative decisions are to be made.
3. When considering whether to temporize or attempt definitive action at intraoperative consultation, current patient condition, procedure morbidity, long-term consequences, and reversibility should all be thoroughly considered.
4. Complex reconstructions, with selected exceptions, are best deferred with temporizing measures until such time that the patient is able to participate in shared decision making.

5. Documentation of intraoperative consultations should be fact-based and include the consultant's primary observations and assessment.
6. When bladder injury is found, it is important to ensure there is no concomitant injury to the ureters or ureteral orifices.
7. When a bladder injury is suspected but not visualized, a cystotomy may be performed to facilitate identification of the area of injury.
8. Early recognition of ureteral injury is important, as failure to do so carries significant morbidity.
9. If ureteral injury is suspected but not visualized, imaging and/or endoscopic assessments should be performed to confirm presence or absence of injury.
10. Consultation for ureteral catheter placement should be reserved for the preoperative setting and for appropriately selected cases.